SIEMENS Healthineers

MAGNETOM

# Function Description System

## Patient Handling

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## 1.1 General

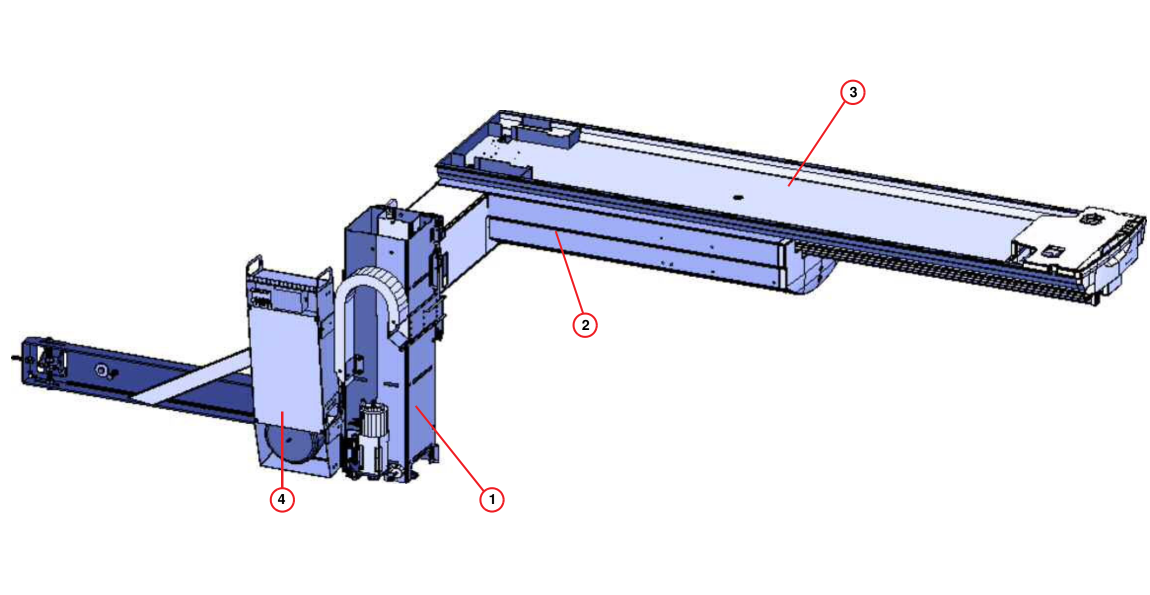
The Patient Handling system is a brand new development, based on the superior design of the Harmony/Symphony components. Many new features were introduced to optimize customer performance. To mention the most important ones: Longer measurement length (e.g. up to 205 cm for MAGNETOM Avanto), faster patient move (Peripheral Angio), support of all Tim applications, wireless Physiological Measurement Unit and many more...

The Patient Handling consists of the following main components:

- **Patient Table** - with a fixed patient tabletop available as a "Standard Bed" with a standard measurement length (e.g. Avanto = 154 cm) or extended travel range (e.g. Avanto with "Tim Whole Body Suite" license = 205 cm)
- **Trolley option** - with Removable Tabletop (RTT)
- **Intercom** - for communication with the patient
- **Physiological Measurement Unit option** - with wireless data transmission of ECG, pulse and respiratory signals and connection possibility for external triggering. Additionally, a PMU display to be mounted to the left front side of the magnet is available.

*Fig. 1: Patient Table, Front View (here: MAGNETOM Avanto)*

*Fig. 2: Patient Table Assemblies: Mechanical View*




(1) Lifting Column
(2) Support Frame
(3) Patient Board
(4) Control Electronics

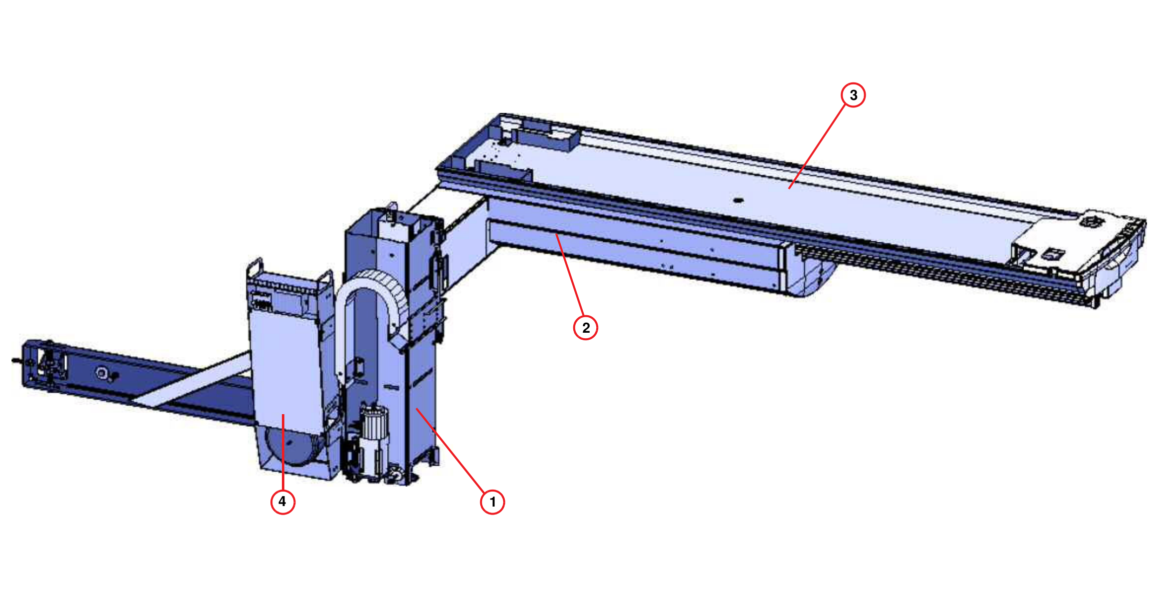
## 2.1 Overview

The patient table is designed as a free floating table, i.e. free foot room for the attending staff and improved access to the patient. A patient weight of up to 200 kg can be lifted. The horizontal travel is supported by a so-called "Telescopic Drive" that translates a relatively small travel range into a wide range to support whole body applications.

Four sub-components are part of the Patient Table:

- **Lifting Column**
- **Support Frame**
- **Patient Board**
- **Control Electronics**

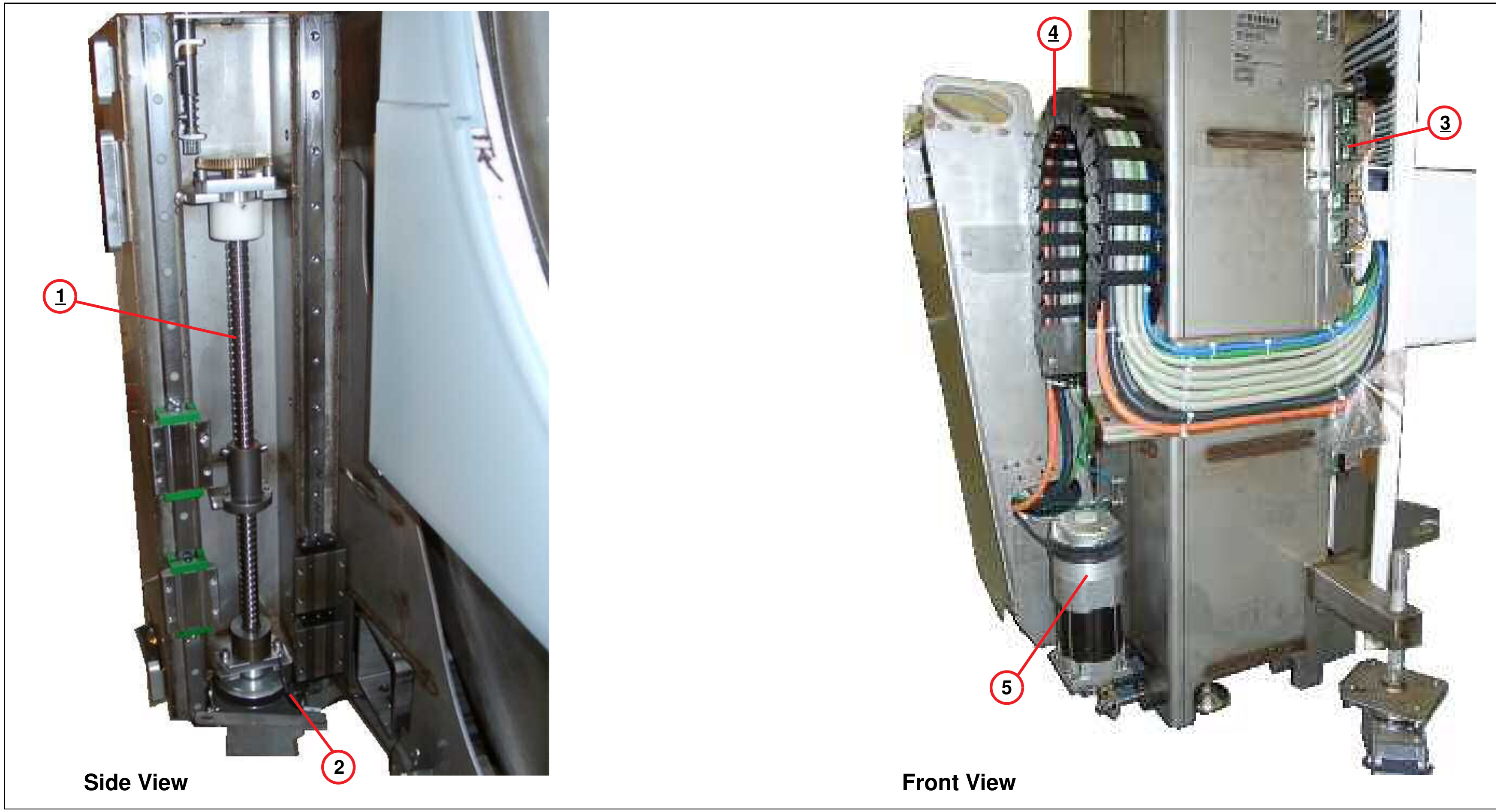
*Fig. 3: Patient Table Assemblies: Mechanical View*




(1) Lifting Column
(2) Support Frame
(3) Patient Board
(4) Control Electronics

## 2.2 Lifting Column

*Fig. 4: Lifting Column Assembly*




(1) Spindle
(2) Spindle-brake temperature sensor
(3) Vertical Switch Assembly
(4) Energy Chain
(5) AC-Synchronous motor

### 2.2.1 General

The lifting column is a free-standing construction with a small overall height and a depth. It is equipped with a well-designed interface to the magnet that can be easily removed or docked if necessary (e.g. in case of transportation problems during new installations). All components can be serviced from the front of the magnet.

Main parts are:

- **Spindle Assembly** - with ball bearing (to minimize mechanical wear) and mechanical brake.
- **AC-synchronous motor** - to drive the spindle via a toothed belt. For safety reasons, the motor contains a second (electro-mechanical) brake (see also "vertical control" description).
- **Vertical Switch Assembly** - to control vertical end positions and speed.
- **Energy Chain** - including control and RF-cables to the patient tabletop.

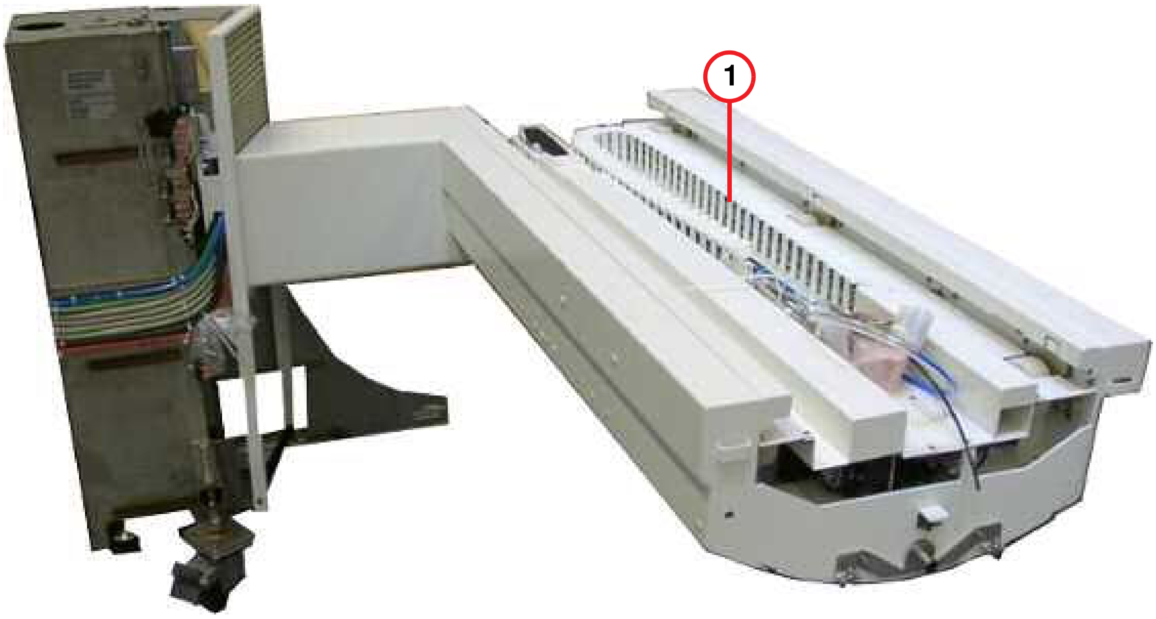
## 2.3 Support Frame

### 2.3.1 General

The Support Frame consists of the following components:

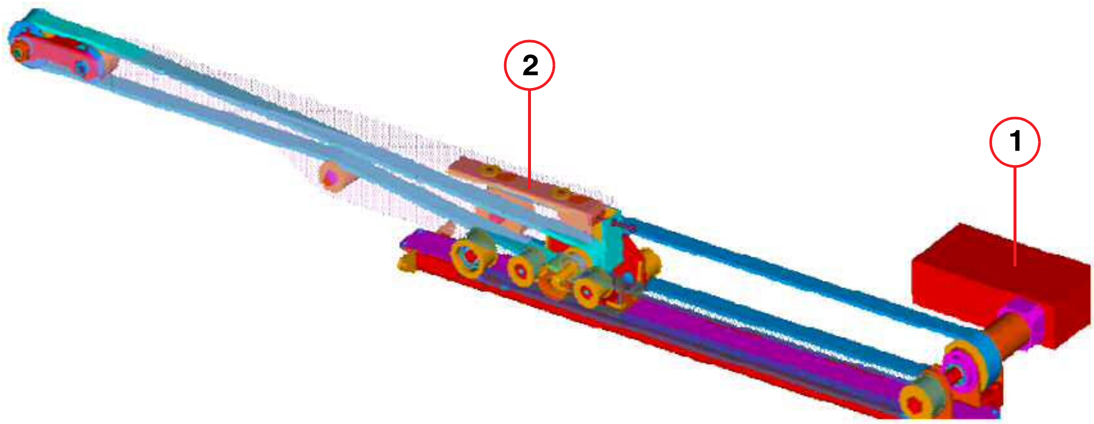
- **Energy Chain** - a cable-routing mechanism feeding all cables to and from the patient board.
- **MDSD** - a horizontal drive that consists of an AC-synchronous motor with an integrated gear-box, position encoder and drive control electronics, all contained in a fully electro-magnetically shielded housing.
- **Alarm Squeeze Bulb Sensor** - (not shown here)
- **Acoustic transducer** - (not shown here)
- **Telescopic Drive** - a new drive mechanism consisting of a drive sleigh and a telescopic outrigger. It allows for an extended tabletop drive range through the magnet. This telescopic drive assembly is driven by the MDSD via a double-sided toothed belt which in turn is driven by a double-sided toothed belt.
- **Sensors** - for horizontal movement and trolley docking (not shown here).

*Fig. 5: Support frame*




(1) Energy Chain

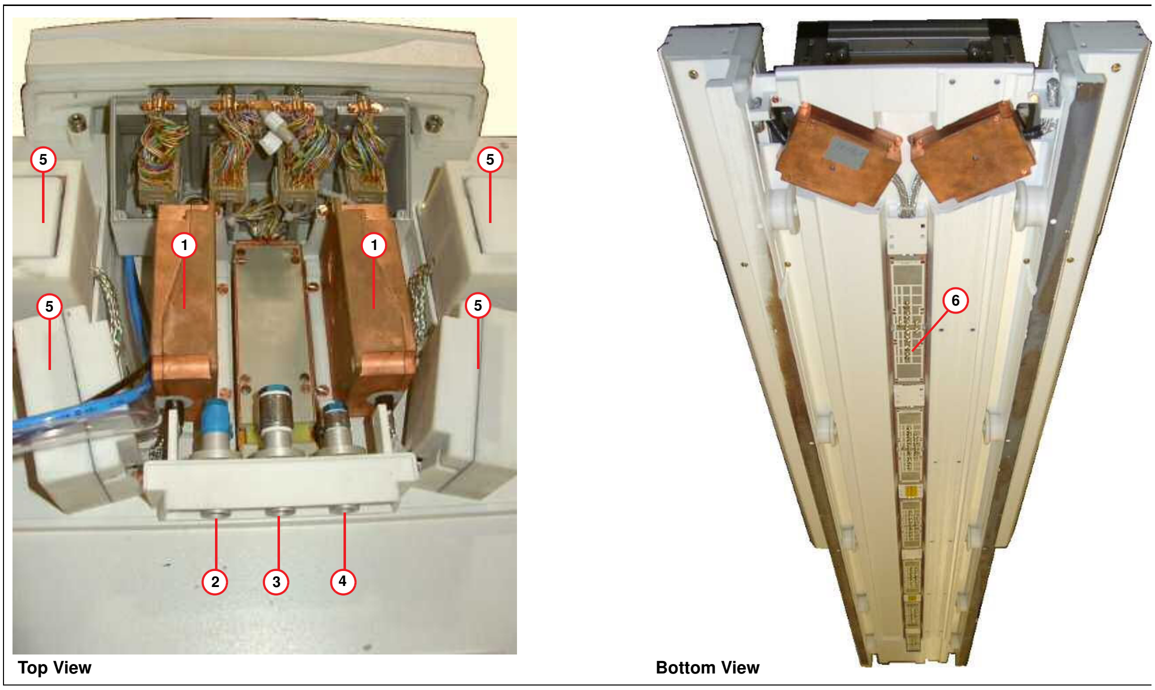
*Fig. 6: Horizontal Drive*




(1) MDSD
(2) Telescopic Drive

## 2.4 Patient Board

*Fig. 7: Patient Board Assembly*




(1) Coil Connector RF-wave traps
(2) Connection for Vacuum Cushions
(3) Connection for head phones
(4) Connection for Alarm Squeeze Bulb
(5) RF-connectors
(6) Integrated RF-wave traps

### 2.4.1 General

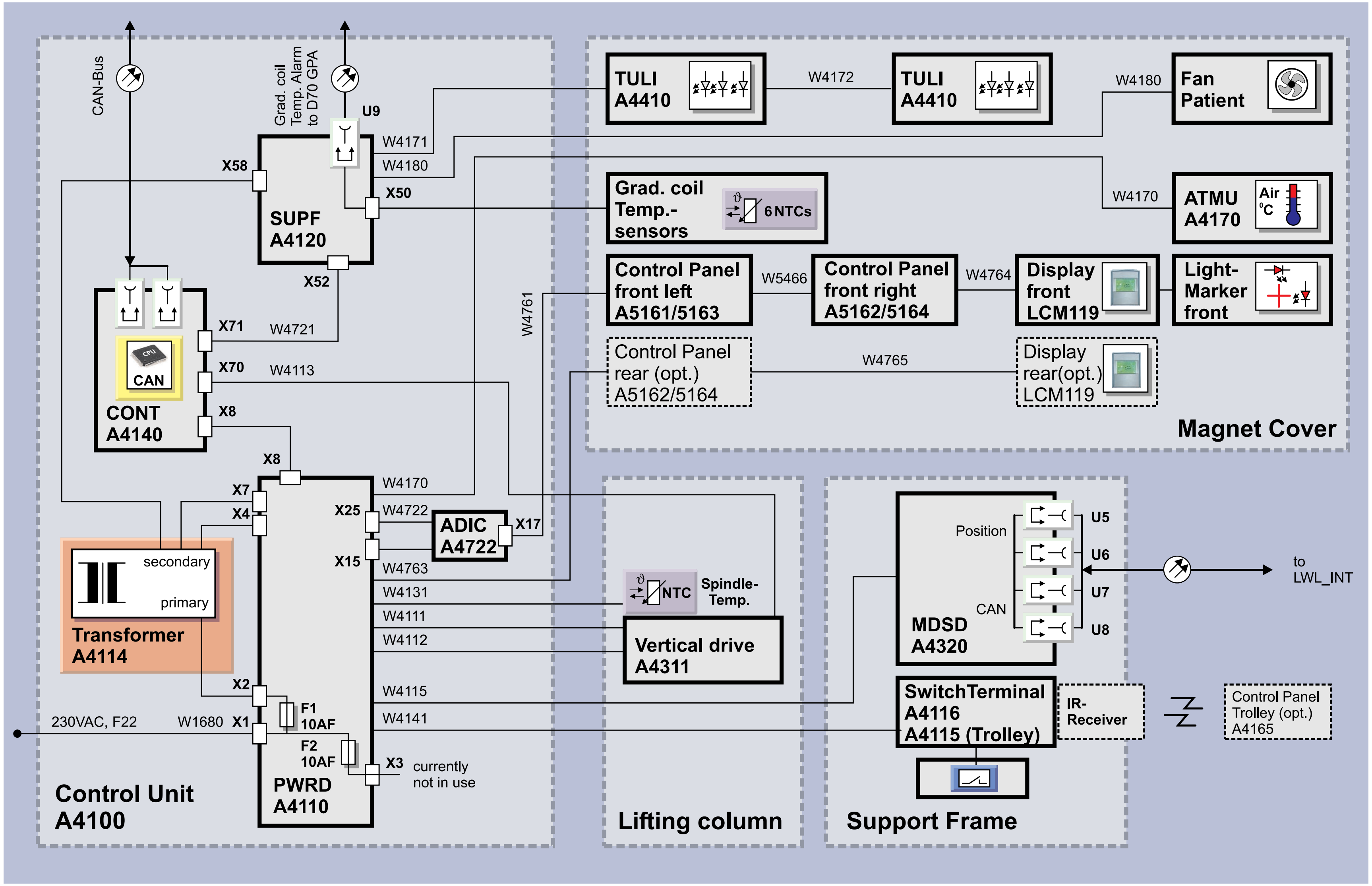
A standard system is equipped with a fixed tabletop. When fitted with the trolley option, it will include an exchangeable tabletop (RTT) that can be separated from the table carrier by using the trolley.

The patient tabletop contains:

- **Coil connector RF-wave traps**
- **Connection for optional vacuum cushions**
- **Connection for headphones**
- **Connection for alarm squeeze bulb**
- **RF-connectors** - for details please refer to the RF-description
- **Integrated RF-wave traps**

## 2.5 Control Electronics

*Fig. 8: Control Electronics Overview Diagram*



### 2.5.1 General

The control electronics consists of:

- **Control Unit A4100** - The assembly box is located in front at the left side of the magnet, close to the lifting column (see overview of patient table). It contains main boards CONT A4140, PWRD A4110, SUPF A4120, (ADIC A4722 for Symphony Tim Upgrades only) and transformer A4114. The Control Unit has to be configured via switches S 1.1 and S 1.2 at the PWRD A4110 (→ Switches / Page 30).

  The controller **CONT A4140** is the communication interface to the CAN bus.

  The **Transformer A4114** and **PWRD A4110** (Power driver) is responsible for low voltage supply, generation of table drive voltages and connection interface to the peripheral components. Symphony Tim Upgrade systems are additionally equipped with an **ADIC A4722** (Adapter Intercom) to adapt the volume adjustment potentiometers of the “old” control panels to the new hardware.

  The **SUPF A4120** (Supplied Functions) is a multi-purpose board for the supply of Patient-Fan, -Light, -Video camera as well as PDAU (see PMU-option) and gradient coil temperature-monitoring.
- **Vertical drive A4311**- Located at the lifting column.
- **MDSD A4320** - Drive for horizontal movement, located inside the support frame.
- **Switch Terminal A4116** (A4115 for trolley option) - To connect switches and sensors to Control Unit A4100. The switch terminal is located inside the support frame.
- **Peripheral components** - Control panels, displays, light markers, tunnel light, patient fan, microphone, acoustic transducer, and Air Temperature Measurement Unit (ATMU).

> i The status of the CONT A4140 software is displayed by LEDs. (Firmware running: yellow LED H72 flashes / Loadware running: green LED H73 flashes).

> i During measurements, CAN bus and CONT A4140 are switched to a “sleep-mode” to avoid RF-interference to the MR-signal.

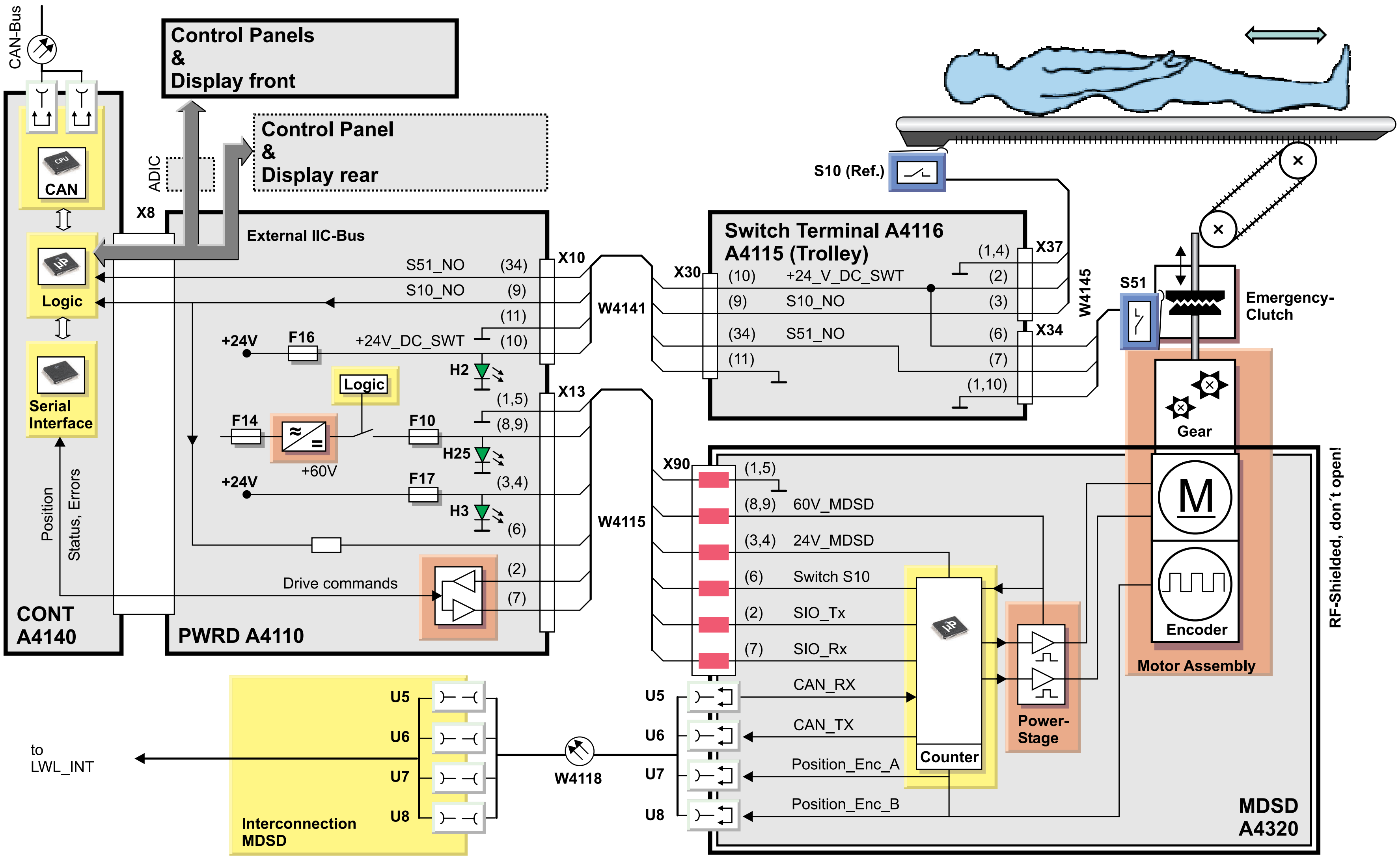
### 2.5.2 Function

The Control Electronics is responsible for the following functions:

- Horizontal Control
- Vertical Control
- Control Panels and Displays
- Emergency Stop Circuit
- Air Temperature Measurement Unit (ATMU)
- Gradient Coil Temperature Monitoring
- Patient Fan, Light and Auxiliary Supply

### 2.5.2.1 Horizontal Control

*Fig. 9: Horizontal Control Overview Diagram*



### Hardware overview

The components involved in the Horizontal Control are:

- **Control Panels**
- **CONT A4140**
- **PWRD A4110**
- **MDSD A4320**
- **Switch Terminal A4116** (A4115 for systems with trolley option)
- **Transformer A4114**

### Horizontal Drive commands

Horizontal movement can be initiated at the control panel or by software CAN-commands. By means of the external IIC-Bus connection, the commands are fed to the CONT A4140, to the PWRD A4110 and via cable W4115 to the MDSD A4320.

The CONT A4140 is connected to the CAN bus to report the patient table status and errors to the system and to drive the table via software (e.g. during Tune-Up procedures). When a measurement is started, the CAN Bus with CONT is set to a “sleep-mode” to avoid RF-interference to the MR-signal.

The MDSD is equipped with a direct CAN bus and position feedback to the AMC (4 fiber optic cables W4118) which is currently used for MDSD configuration and status only. Incorrectly connected cables at this point may result in a CAN1 bus blockage.

### Horizontal Power Stage

The MDSD driver circuit converts horizontal movement commands into pulse width-modulated signals, amplified by an on-board Power Stage. The driver is supplied with 24 VDC by the PWRD (green LED H3, Fuse F17). The 60 VDC Power Stage supply voltage is switched off by a logic (green LED H25, Fuse F10) in case of an emergency (Table Stop). A DC motor with integrated relative position encoder and gearbox is part of the MDSD and drives a double-sided toothed belt - coupled to a drive sleigh and a telescope outrigger.

**Position Detection**

Every time when the power is removed from the patient table, its absolute horizontal table position gets lost. In order to recalibrate the absolute horizontal position, the table has to be moved to the home position (switch S10 closed). From this reference position, any offset can be calculated, counting the increments of the MDSD position encoder.

The MDSD remains active during measurements, therefore the closed-loop position detection and motor control circuit can lock the patient board during measurement.

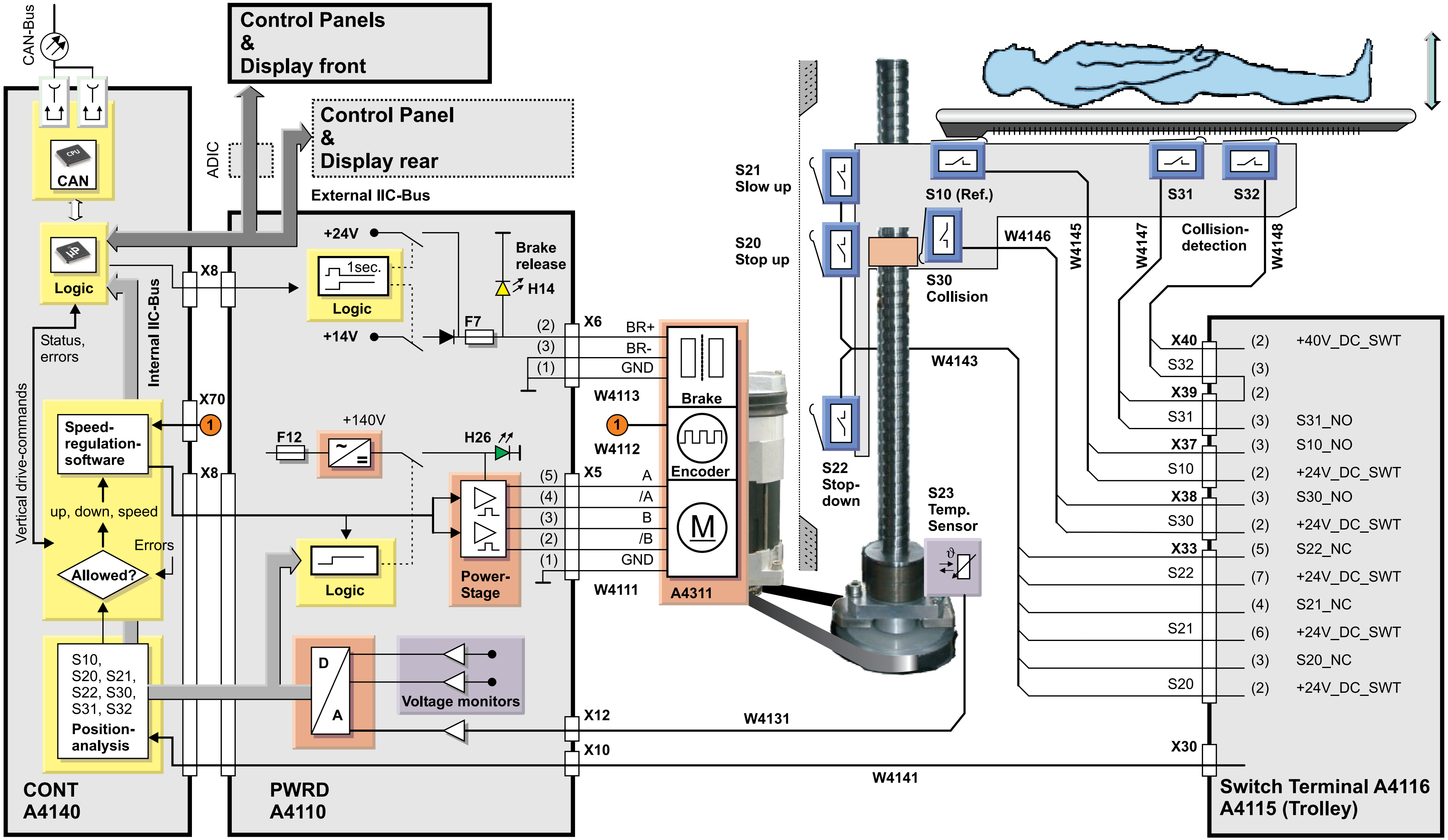
> i The MDSD is shielded to avoid RF-interference to the MR-signal. In case of failure, replace the whole unit, don’t try to open the box.

### Emergency Clutch

In case of emergency, the operator can release a mechanical clutch to decouple the patient board from the horizontal drive mechanics - now the patient can be easily pulled out of the bore by hand. The clutch is monitored by Switch S51 (see description switches).

### 2.5.2.2 Vertical Control

*Fig. 10: Vertical Control Overview Diagram*



### Hardware overview

The components **involved** in vertical movement are:

- **Control Panels**
- **CONT A4140**
- **PWRD A4110**
- **Vertical Drive A4311** (vertical motor assembly)
- **Switch Terminal A4116** (A4115 in case of installed trolley option)
- **Transformer A4114**

### Vertical Drive commands

Vertical move commands from the control panels are fed to the CONT A4140 by means of an external IIC bus connection. Before movement, the CONT analyzes if the table is in a position where vertical movement is allowed and checks for errors.

If everything is ok, the vertical drive commands are processed by a speed regulation software and the nominal current values are routed to the PWRD A4110 via connector X8.

### Vertical Power Stage, Speed regulation

The PWRD A4110 is responsible for converting the drive commands into pulse width-modulated nominal current values, amplified by a 140 VDC Power Stage. A green LED (H26) indicates voltage present.

The Vertical Drive A4311 consists of an AC-synchronous motor with integrated relative position encoder and integrated brake that is released electrically. The position encoder signals are routed via cable connection W4112 to the CONT board, where they are used for detection and regulation of vertical movement speed.

### Vertical Brake

The mechanical friction of the ball-bearing spindle is not sufficient to safely lock the vertical position. To avoid safety hazards, two independent vertical brakes are implemented:

- **Mechanical brake** - at the bottom side of the spindle
- **Electromechanical brake** - part of vertical drive A4311

The mechanical spindle brake engages during downward movement only. It has no friction in the opposite direction. During excessive table travel, the brake may overheat (won´t happen during normal operation). An attached sensor reports the actual brake temperature via cable connection W4131 to the PWRD A4110 where it is digitized and reported to the CONT A4140 via the internal IIC bus.

In case of overheating, a temperature warning is generated and sent via CAN bus to the system. If excessive travel continues, the CONT disables the "down" direction and reports a temperature alarm. In this case, the patient table can still be moved up fully, during the following examination the brake will cool down again.

The electro-mechanical brake locks if its power is removed. When the vertical motor is activated, a logic at the PWRD A4110 switches on 24 VDC to release the brake fully. After one second, the logic removes the voltage and switches on 14 VDC to avoid brake overheating. A yellow LED (H14) at the PWRD indicates the brake status. In case of electrical overload, fuse F7 trips off.

#### Vertical Collision Detection

If vertical down movement is blocked by an obstacle and the Support Frame gets lifted from the spindle nut, switch S30 will open and block any further downward movement. A blockage that tilts the patient board will be sensed by switch S31 and S32 (connected in series). An error message is generated, but table up movement is still possible to easily remove the obstacle.

> For service purposes, the patient table can be moved vertically using the hand crank (stored under the top cover of the lifting column ).

### 2.5.2.3 Control Panels and Displays

#### Hardware overview

The operator can control and monitor the patient table functions by:

- **Control panel A5161/5163** - front, left side
- **Control panel A5162/5164** - front, right side
- **Display LCM119** - front, with Laser Light Localizer
- **Optional Back Control Panel A5162/5164**
- **Optional Back Display LCM119**

At the control panels, patient table motion is controlled by joystick and push buttons. Additionally, the lighting, air flow and the physician's voice/background music volume can be adjusted to the patient's needs.

To avoid interference with the MR signal, the CONT, IIC bus, the Control Panels and Displays are switched to a so-called “sleep-mode” during measurement. The hardware can be reactivated by the /WAKE_PAN signal that is sent to the CONT board if movement keys are pressed.

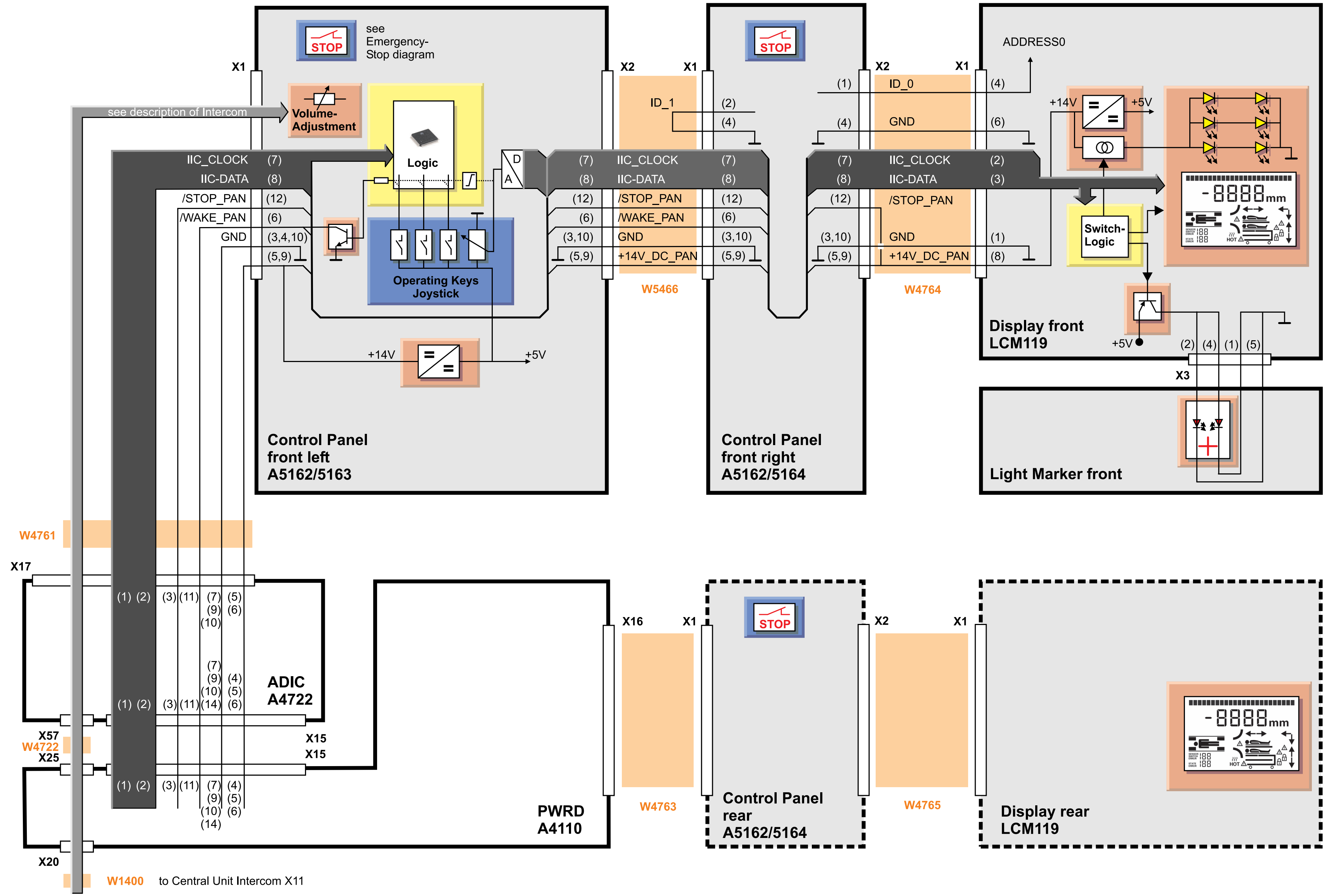
#### IIC bus addressing

The Control Panels and Displays are connected via the external IIC bus and the PWRD A4110 to the CONT A4140. They contain identical circuits and are not equipped with onboard addresses. The CONT can identify connected panels and displays during power-up by reading the address that is hard-wired by the ID_0 and ID_1 signals.

For troubleshooting, the panels and displays can be swapped or connected directly to the PWRD A4110.

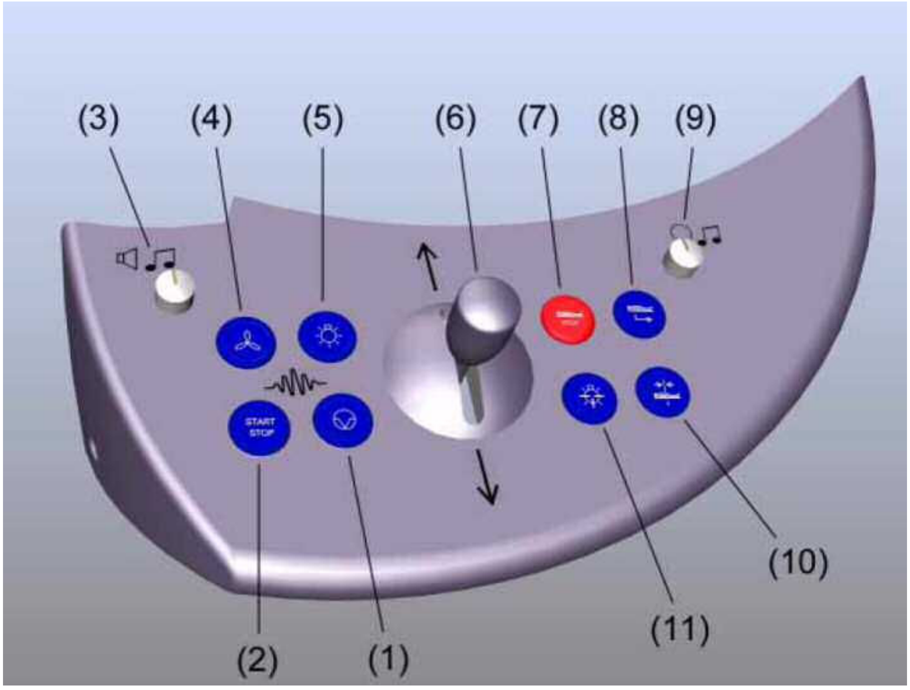
 

*Fig. 11: Control Panels and Displays*



### Control Panel keys

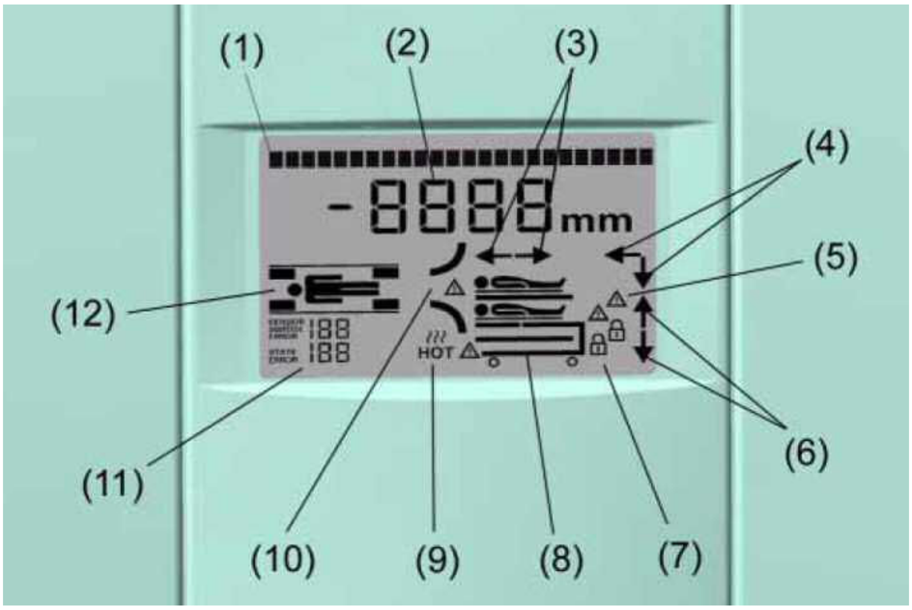
*Fig. 12: Control Panel keys*




(1) Not used
(2) Measurement Start/Stop button
(3) Wall speaker volume control
(4) Tunnel ventilation button
(5) Tunnel lighting button
(6) Joystick table movement
(7) Table Stop button
(8) Home position button
(9) Headphones volume control
(10) Center position button
(11) Laser light localizer button

## Display graphics

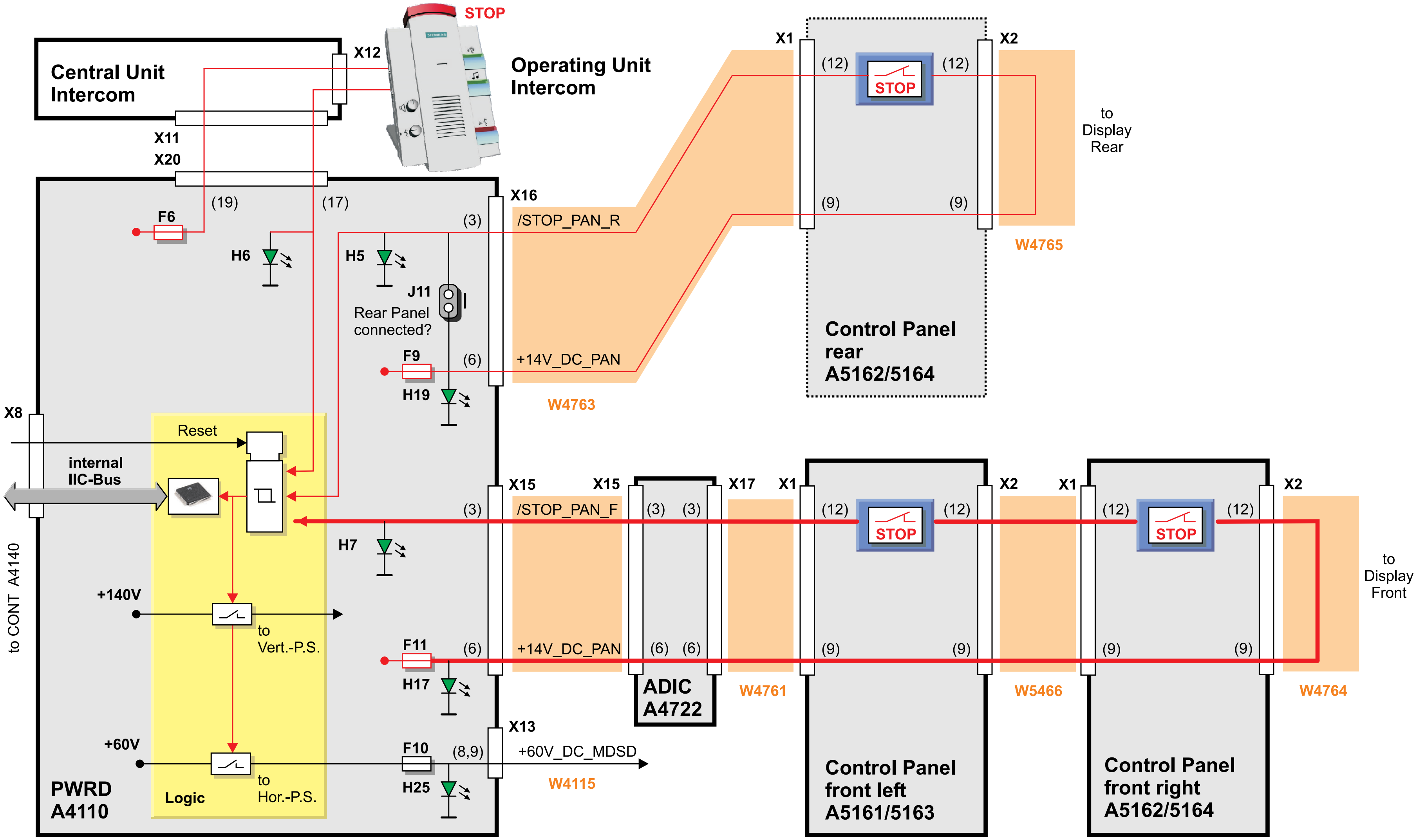
*Fig. 13: Display*




(1) Text output line for status of patient comfort devices such as fan, light, and intercom volume
(2) Actual horizontal table position
(3) Direction arrows for table movement
(4) Combined arrows (vertical, horizontal)
(5) Collisions when moving the table in vertical direction
(6) Direction arrows for table movement
(7) Trolley locked
(8) Trolley docked
(9) Overheating of spindle brake
(10) Patient on table, Patient Table ok - Symbol flashes in case of severe errors)
(11) For service: status of switches, sensors, errors and SW-states
(12) Coil socket assignments

## 2.5.2.4 Emergency Stop Circuit

*Fig. 14: Emergency Stop Circuit*



### Hardware overview

The patient table is equipped with up to four stop buttons found at these locations:

- **Two front Control Panels**
- **Back control panel (option)**
- **Intercom Operating Unit**

### Stop circuits

Each circuit uses the +14 VDC (+14V_DC_PAN) panel operating voltage, generated at the PWRD A4110 and fed there via fuses (F6, F9, F11). Two green LEDs (H17, H19) indicate the presence of the +14V_DC_PAN supply for the front and optional back panel circuitry.

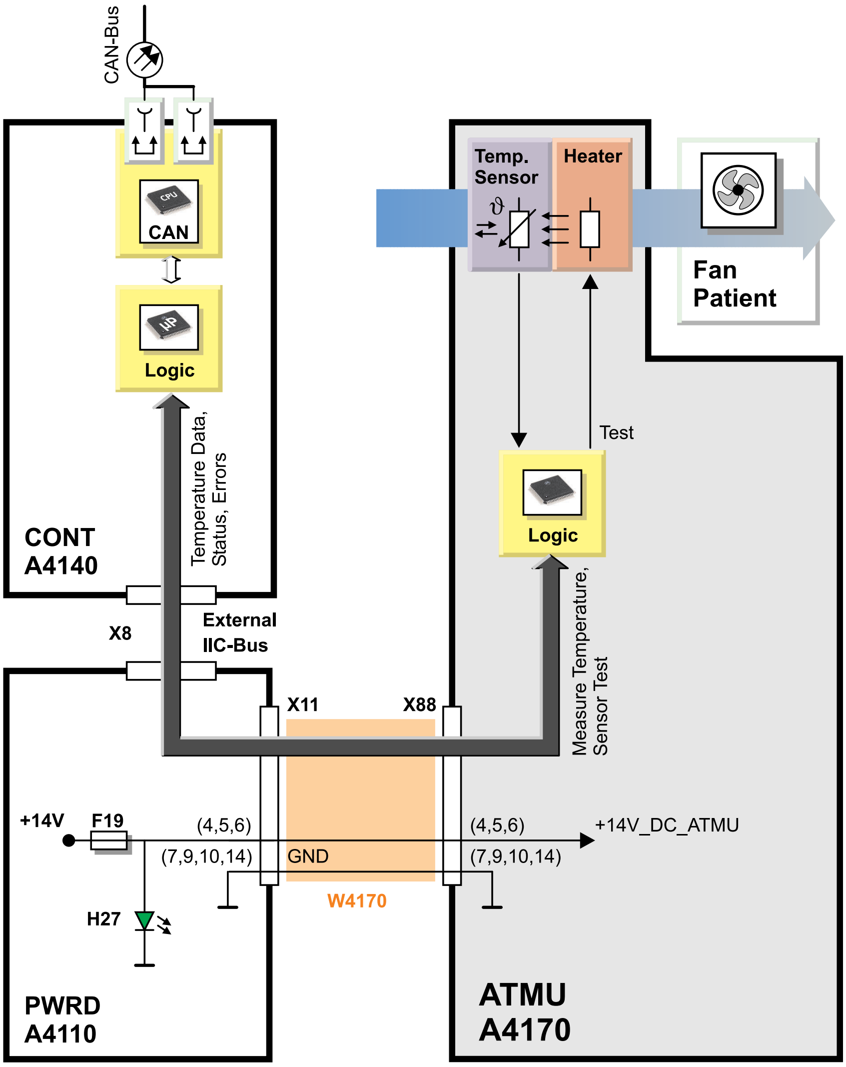
The voltage is routed back via the Stop buttons (closed contacts) to the PWRD A4110, indicated by LEDs H5, H6 and H7 that light up. Pressing one of the buttons causes an interrupt, to be detected by an error flip-flop at the PWRD that disables the +140 VDC supply for the Vertical Power Stage and the +60 VDC for the MDSD Horizontal Drive. Additionally, the stop condition is reported to syngo MR via CONT A4140 and the CAN bus.

To reset the Table stop, the joystick at one of the Control Panels has to be deflected in both directions in one sequence.

Jumper J11 must be fitted if the Back Control Panel A5162/5164 is not installed. If the Back Panel exists, the jumper must be removed.

### 2.5.2.5 ATMU

*Fig. 15: ATMU Block Diagram*




#### Background

Prior to starting a measurement, the RF Safety Watchdog (RFSWD) software calculates the SAR look-ahead parameters. Since the maximum allowed RF exposition to the patient also depends on the ambient air temperature, it has to be measured. This is achieved by the Patient Environment Temperature sensor of the ATMU (Air Temperature Measurement Unit), located at the air inlet of the patient fan.

#### Air Temperature Monitoring

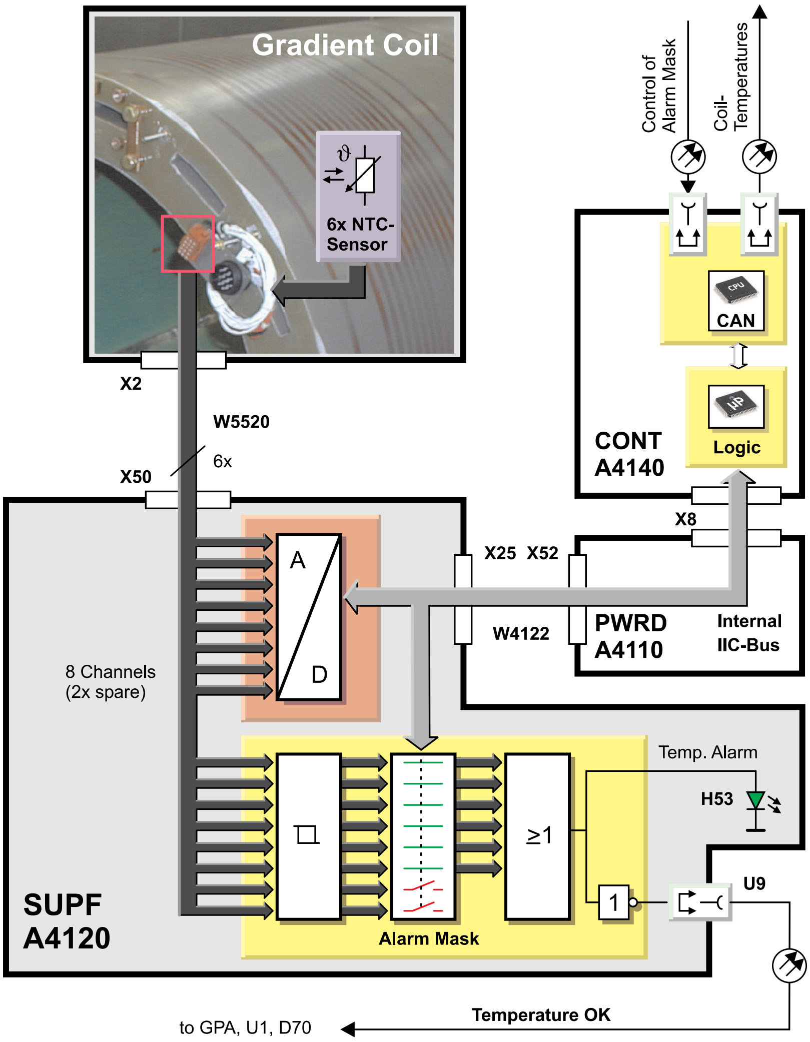
Via the AMC and CAN_Bus connection, a request is sent to the CONT A4140 to read the actual air temperature at the ATMU A4170. The actual temperature measured is reported back to the RFSWD.

### Temperature Sensor Check

The ATMU Temperature Sensor can be tested in the Service Software / QA (see Service Routines).

## 2.5.2.6 Gradient Coil Temperature Monitoring

*Fig. 16: Gradient Coil Temperature Monitoring*




### Gradient Coil Temperature measurement

The temperatures of the gradient coil windings are monitored by the SUPF A4120 board. The gradient coil has six NTC resistors, one located on each of its six windings which are fed to a comparator logic on the SUPF A4120 for evaluation. (At the Espree Slim Line Gradient Coil AS 022SL, 4 additional redundant NTCs are part of the coil and can be connected instead of broken resistors.)

If the gradient coil temperature exceeds the limits, the GPA will be switched off by removing the **Temperature OK** sum signal (FOC U9 SUPF to U1 D70 at the GPA. (Light on = all temperatures are within limits).

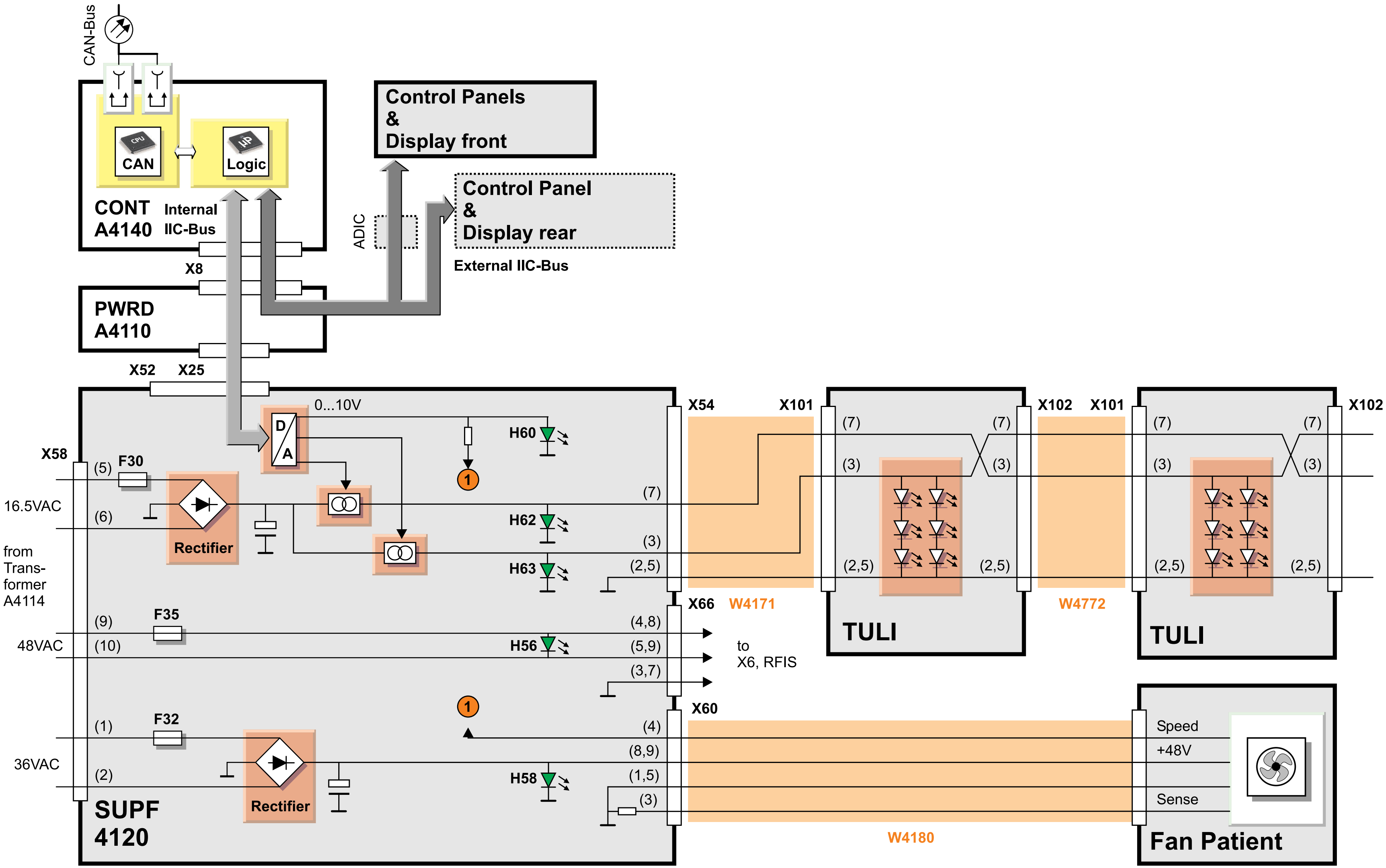
Additionally, the temperature values are fed via internal IIC-Bus connection to the CONT A4140 where they can be read on the CAN-Bus by the host computer.

### Alarm Mask

The SUPF temperature detection logic is equipped with eight monitoring channels of which six are currently used to measure the gradient coil temperature (6 NTC's). The two unused channels are masked out by an alarm mask logic loaded by the CAN-Bus.

### 2.5.2.7 Patient Fan, Light, and Auxiliary Supply

*Fig. 17: Patient Fan, Tunnel Light and Auxiliary Supply*



### Patient Fan

For increased patient comfort a cooling fan, having three fan speeds, can be adjusted at the Control Panels or via syngo MR. The commands are routed from CONT A4140 to PWRD A4110 and SUPF A4120 via the internal IIC bus connection. A D/A converter at the SUPF feeds out the control voltage (0-10 V, indicated by LED H60) for the fans' speed control circuit. The fan operating voltage is supplied by transformer A4114 (36 VAC), rectified on board and indicated by LED H58 (FAN_DC_PAT, 48 VDC).

### Tunnel Light (TULI)

Similar to the patient fan, the brightness of the two back LED tunnel lights is adjustable in steps.

The D/A converter at the SUPF sets up two independent brightness current control circuits, supplied by a rectified 16.5 VAC voltage. LED H62 (for TULI 2) and H63 (for TULI 1) indicate the presence of each supply.

### Auxiliary Supply

An auxiliary supply voltage (48 VAC) is taken from transformer A4114 and fed to connector X6 of the RFIS for generating the -450 VDC static-detuning voltage for the body coil. If the voltage is present, LED H56 lights up.

## 2.5.3 Switches

i For a description of switches, please refer to the Troubleshooting Guide.

## 2.5.4 Jumpers

Tab. 1 Summary, jumpers

| Jumper | Location | Default position | Description |
|---|---|---|---|
| J1 | CONT A4-140 | Jumper fitted | Automatic starting of CONT A4140 loadware after power up. LED H73 blinks when loadware is running (except in sleep-mode) |
| J11 | PWRD A4110, close to X16 | Jumper fitted | Must be fitted if the back Control Panel A4162 option is not installed. Closes Table Stop circuit. If installed, the jumper must be removed; otherwise, the back Stop button won´t work. |
| J12 | PWRD A4110, close to X14 | Jumper fitted | Must be fitted, if the Operational Table option is not installed. Closes Table Stop circuit. |

### 2.5.5 Service routines

Since the CONT A4140 is equipped with a built-in “Service Mode”, you will not find functional tests in the Test Tools under SESO (only a communication and status test and a temperature sensor check in QA).

#### 2.5.5.1 Display of current errors

The table control is able to report errors in the following ways:

- **Via CAN bus to the host** - Here, the error ID´s are identified and connected to the corresponding error text stored in syngo MR.
- **Numerically at the patient table display ("Error: nnn")** - In case of multiple errors, the display cycles among them sequentially. The number displayed corresponds to the error ID in the eventlog and TSG.
- **Error text at the patient table display** - If there has been no CAN host communication since the last power-up, the error text is displayed in the text area.

#### 2.5.5.2 Reset of table movement errors

To reset table movement errors (e.g. "Table Stop" was pressed), the joystick has to be deflected in both directions once in sequence.

#### 2.5.5.3 Service Mode

An extremely useful tool for patient table troubleshooting and tests is the so-called "Service Mode". This mode can be invoked by pressing a special key combination at one of the control panels (see TSG). Tests are available for:

- Switches and sensors
- Operating panels
- Displays (including contrast adjustment)
- Temperatures (spindle brake, PWRD and SUPF)

> During active service mode, motor-driven table movements are not possible. For a detailed description of the service mode, please refer to the TSG.

#### 2.5.5.4 Debug display

The "Debug display" allows internal software "states" to be reported during normal table operation. In case of severe problems, these states can help the HSC or Lab in solving the problem. When the "Debug display" is activated, you will find the following information at the left bottom corner:

- Display of the state number
- Display of switch status

### 2.5.5.5 Debug marking

Debug marking generates a pseudo error (no real error!), shown as Message ID 191, "Marking current situation as notable by user" in the eventlog. Setting such a reference mark and provoking a potential error directly afterwards may facilitate navigation through the error messages.

### 2.5.5.6 Temperature Sensor Check

The ATMU Temperature Sensor can be tested in the Service Software / QA.

When the test is started, the CONT A4140 switches off the patient fan and sends a test request to the ATMU where a heater resistor, mounted close to the sensor, is energized. The Controller software checks twice in a time frame of 30 seconds for a temperature increase of 5 degrees centigrade. The test will fail if no temperature increase is detected or if the sensor temperature prior to starting the test exceeds 35 degrees centigrade. Finally, the atient fan is switched on again.

After repeated tests, the user should allow the sensor to cool down properly (approximately 15 minutes). Otherwise, the tests may fail.

## 3.1 Overview

To facilitate patient setup, the system can be equipped with an exchangeable tabletop (RTT) to be separated from the table carrier with the help of a movable trolley. The essential differences with respect to the standard patient table are:

- **Exchangeable TableTop (RTT)**- Can be separated from the table carrier with the trolley.
- **Trolley** - Movable tabletop transport device, equipped with an IR remote control to drive the patient table in the vertical direction.
- **Automatic Plug Connection (APC)** - Special plug in-between RTT and table carrier to connect the RTT switches to the table electronics. Opens automatically, if the RTT is separated from the table carrier.
- **Switch terminal RTT A4115** - Distribution board for table carrier/RTT control switches. Contains an IR receiver for the trolley remote control.

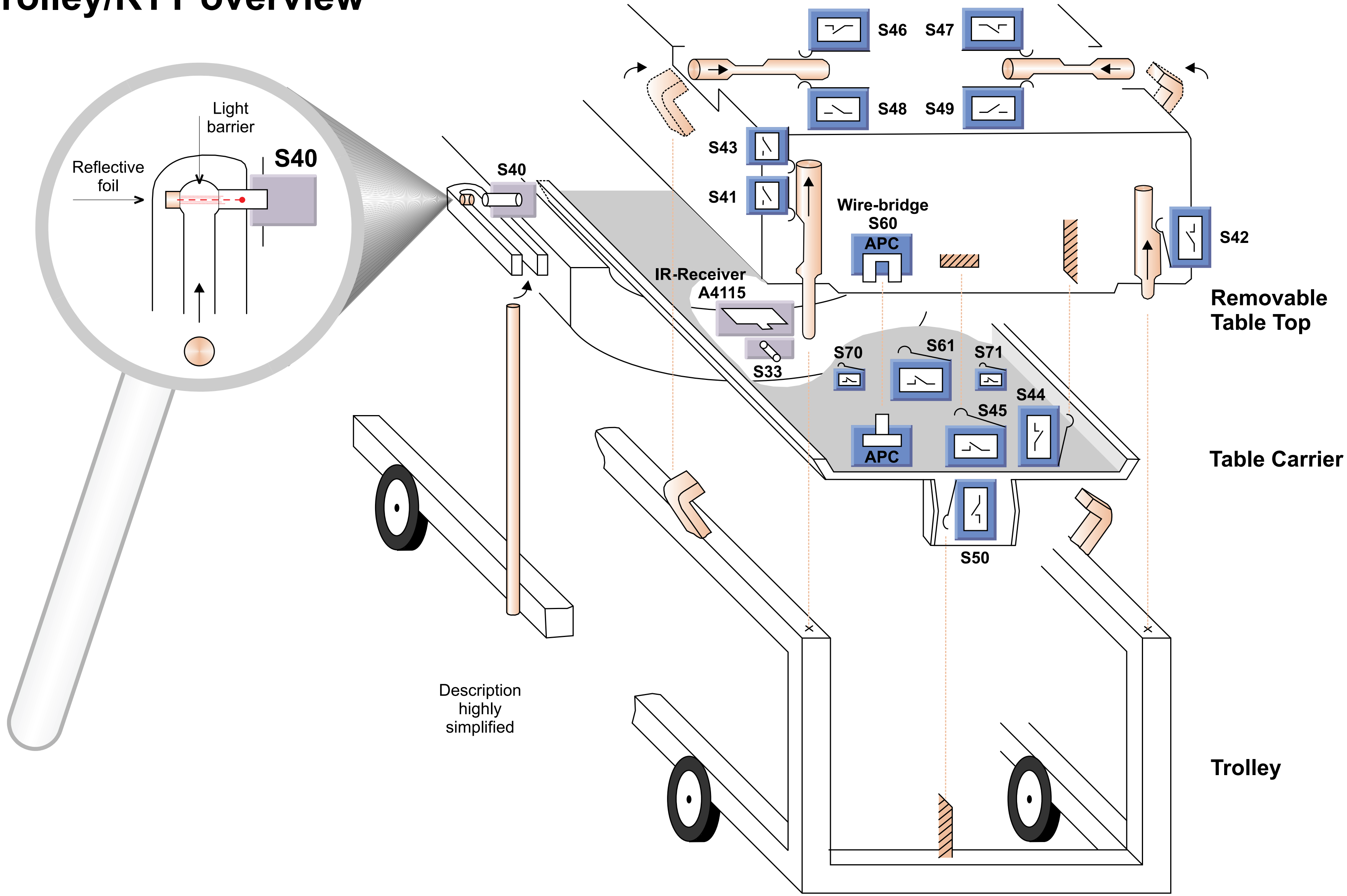
The Trolley/RTT option is system specific, even though the hardware looks identical. Never change parts of the Trolley between different system types, e.g. Avanto and Espree.

The Trolley/RTT option must be enabled by a configuration switch S1.1 at PWRD A4110.

Upgrades of a standard system with the trolley/RTT option has to be performed by specially trained personnel (e.g. Hegele).

*Fig. 18: Trolley/RTT Overview*

# Trolley/RTT overview



## 3.2 Trolley/RTT transfer process

### 3.2.1 Function

The trolley/RTT operation is monitored and controlled by a set of additional switches and sensors located at the RTT, table carrier and support frame.

#### 3.2.1.1 RTT transfer from the table carrier to the trolley

- **Table down movement, high speed** - When S41 is released, vertical travel is switched to slow speed.
- **Table slow-down** - As soon as S42 is actuated or S43 is released, down movement is stopped. A minor residual movement is possible, however, the Automatic Plug Connection (APC) must not open at this point in time.
- **Engaging RTT locking mechanism (foot pedal)** - Operation of the foot pedal will mechanically lock the RTT to the trolley. Both the trolley hooks must capture the RTT safely, monitored by S46 - S49. All RF coil plugs and air hose connectors need to be disconnected by the operator before further down movement is possible (see description in the following pages).
- **Table down movement** - The table can be lowered down further, S45 detects the separation of RTT from the table carrier and the APC will open.

  When reaching the lower vertical end position (valid for trolley operation only), S50 is actuated and the down movement stops. Minor residual movement is possible.
- **RTT can be transported on trolley**

> RTT transfer to the trolley requires both trolley hooks to be fully engaged/locked.

#### 3.2.1.2 RTT transfer from trolley to table carrier

- **Table up movement, high speed** - When S44 is actuated, vertical travel is switched to slow speed.
- **Table moving up slowly** - As soon as S45 is actuated, the up movement is stopped. A minor residual movement is possible, the APC must be fully connected at this point in time.
- **Releasing the RTT locking mechanism (foot pedal)** - Both trolley hooks must be fully released and monitored by S46 - S49. Further up movement will be enabled.
- **Table up movement** - When the RTT is elevated off the trolley frame, S42 is released and S43 is actuated. Vertical travel to the upper end position is possible.

> RTT transfer back to the table carrier requires that both trolley hooks are fully released/ unlocked.

### 3.2.1.3 Optical Sensors

Two optical sensors are responsible for a safe, collision-free RTT transfer process.

- **Trolley docking sensor S40 (docking bracket)** - Correct docking of the trolley with respect to the patient table is monitored continuously via light barrier S40. When correctly docked, the stainless steel trolley guide bar interrupts the optical connection between light barrier S40 and the reflective foil, mounted at the opposite inner side of the docking bracket (docked=high sensor output level). The +24 V OUT_DC_SWITCHED voltage is switched off. The collision detection sensor S33 is no longer operational. When the trolley is removed or docked incorrectly, the light of sensor S40 is fully mirrored back by the reflective foil, the sensor output changes to low status and the collision detection sensor S33 is switched on (OUT_DC_SWITCHED voltage = +24 VDC)
- **Collision detection sensor S33 (table foot end)** - Any object blocking the detection range of optical sensor S33 will be recognized by the table electronics; the vertical movement will be blocked (blockage=low output level).

The collision detection sensor S33 is switched off when the trolley is docked correctly (detected by S40).

The sensors contain a potentiometer that is adjusted to maximum sensitivity by factory. In exceptional cases (e.g. sensor malfunction caused by reflective objects) a readjustment might be neccessary.

### 3.2.1.4 Removal of coil plugs

To avoid mechanical damage, the operator must remove all connected coil plugs prior to the RTT transfer from the table carrier to the trolley.

Currently (as of 03/05) the table electronics are not able to identify remaining plugs except for the two "horizontal" sockets X7 and X9 at the foot end of the table. Switches S70 and S71 will detect if the coil plug cover is open.

To avoid damage to the plugs, decoupling of the RTT from the table carrier is blocked and a reminder message appears on the table display. After you acknowledge this with the "Speed" button (Control Panel), the RTT can be decoupled.

In future software versions, a coil-code check will be implemented, making sure that none of the coil plugs has been forgotten.

### 3.2.1.5 Removal of air hose connectors

The following air hose connectors must be removed prior to RTT transfer from the table carrier to the trolley.

- Connector for vacuum cushions
- Connector for headphones
- Connector for alarm squeeze bulb

Plugged-in connectors are detected by switch S61 at the table carrier (connector plugged in = low signal). RTT decoupling will be blocked.

### 3.2.1.6 IR remote control

The trolley is equipped with an infrared remote control to drive the patient table in the vertical direction. The IR sensor is located at the foot end of the table at the switch terminal RTT board A4115.

## 3.2.2 Switches

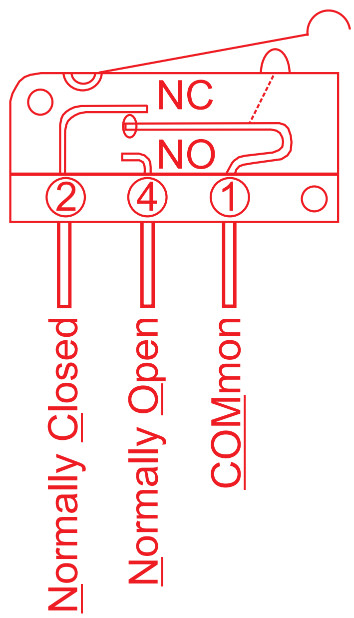
### 3.2.2.1 Plausibility check of switch conditions

The plausibility of the logic status of numerous trolley/RTT switches and sensors is checked by the table electronics. In case of deviations from the predefined status, a table stop is generated and the respective error message is sent to the host.

Some specific switch errors cannot be detected by the plausibility check. They can be recognized by typical table malfunctions (see TSG).

### 3.2.2.2 Microswitches

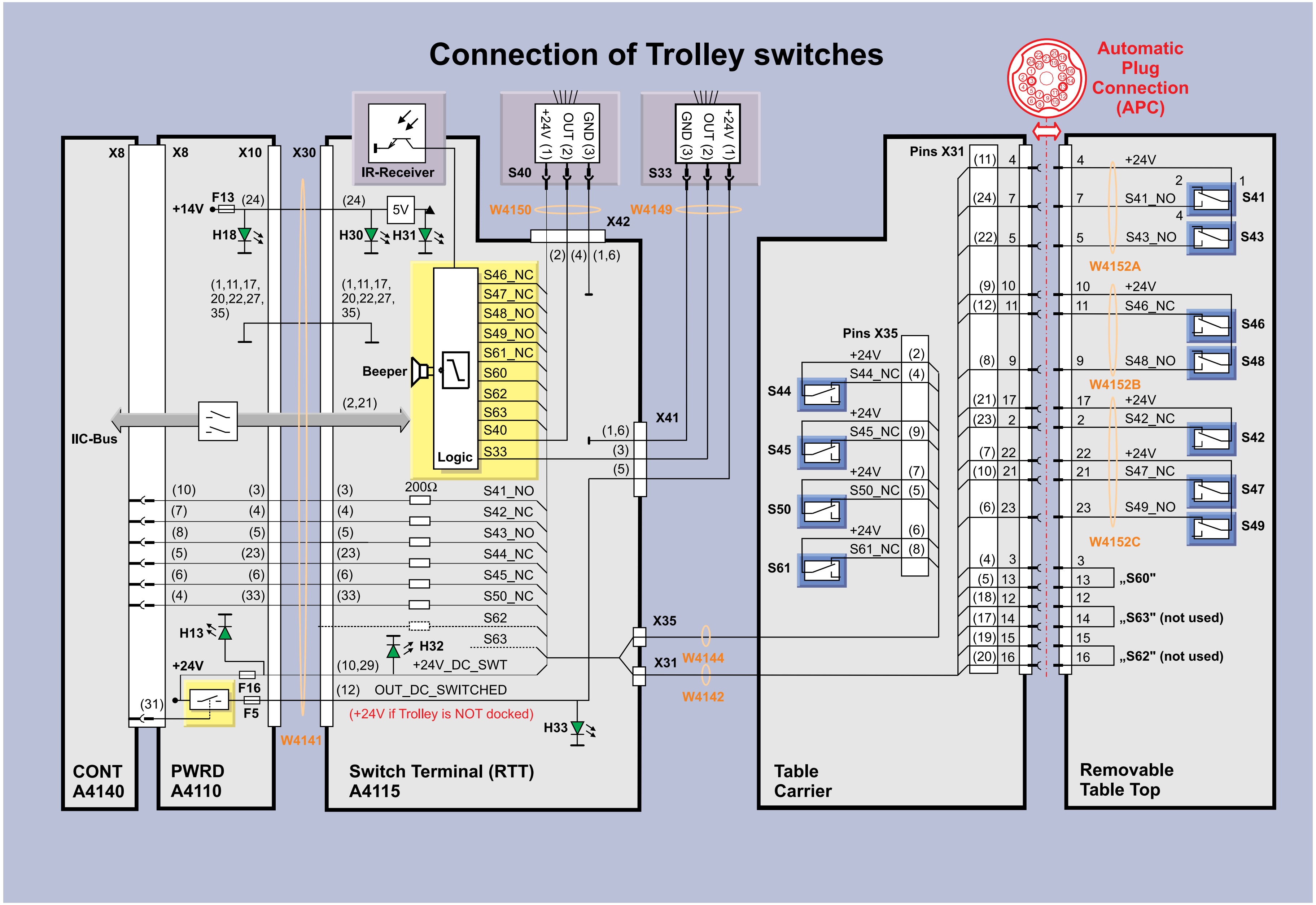
*Fig. 19: Microswitches*




### 3.2.2.3 Connection of Trolley Switches

(see next diagram)

*Fig. 20: Trolley switches/sensors*

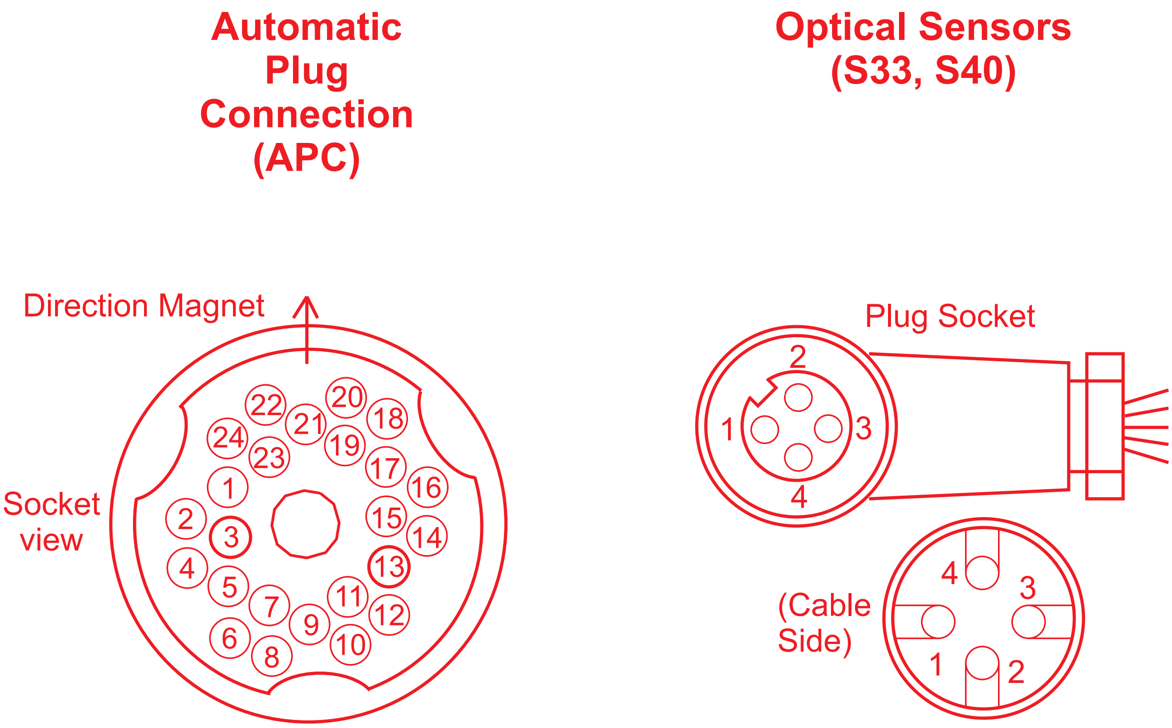


#### 3.2.2.4 List of Trolley/RTT Switches

For a description of switches, please refer to the Troubleshooting Guide

### 3.2.3 Connectors

*Fig. 21: APC and optical sensors*



## 4.1 Overview

Some cardiological and abdominal imaging techniques require physiological triggering for improved image quality and increased diagnostic value of the images. For example, when running a heart study it is very important to know, whether specific images have been measured during the diastolic phase (heart muscle relaxed) or during the systolic phase (heart muscle contracted). The system control unit which executes the measurement (AMC), uses the physiological triggering signals for acquiring image data according to the desired physiological state.

Physiological monitoring is also used to monitor the patient’s well-being.

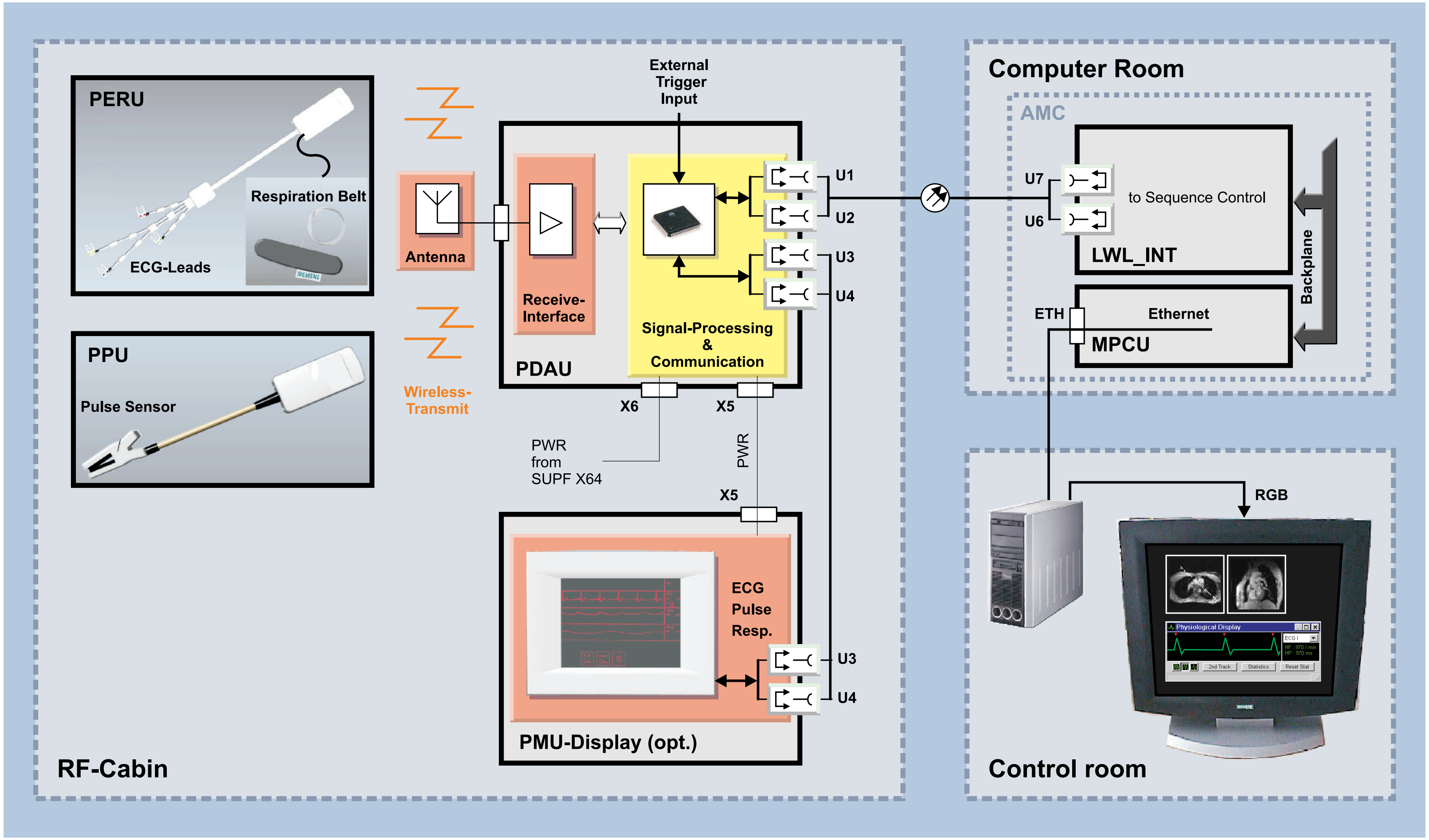
Physiological signals available for triggering:

- **Electro-Cardio Gating** - ECG
- **Respiratory** - chest movement
- **Pulse** - heart rate
- **External triggering source** - TTL input

The PMU consists of the following components:

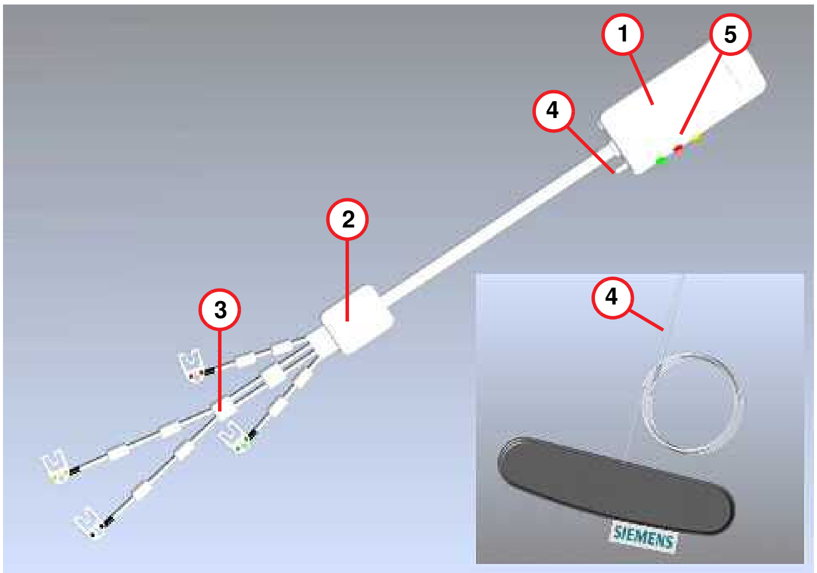
- **PERU** - a wireless Physiological ECG and Respiratory Unit
- **PPU** - a wireless Peripheral Pulse Unit sensor
- **Antenna** - to receive the PERU and PPU signals. Located at the magnet behind the front funnel
- **PDAU** - the Physiological Data Acquisition Unit receives and processes the triggering signals from the sensors
- **PMU display (option)** - a monitor for displaying the physiological signals at the magnet

*Fig. 22: PMU Overview*



## 4.2 PERU

*Fig. 23: PERU*




(1) Transmitter
(2) Preamplifier
(3) ECG leads
(4) Connection for Respiratory belt
(5) Status LEDs

### 4.2.1 Function

The PERU (Physiological ECG and Respiratory Unit) is able to simultaneously record two ECG signals and one respiratory signal (detected by a connected belt). Both signals are digitized and transmitted wirelessly using the Bluetooth standard (2.472 GHz bi-directional). Maximum transmit power is 1 mW with a usable transmission radius of about 3 meters.

The PERU is supplied by an integrated lithium battery, rechargable in less than 3 hours.

### 4.2.2 LEDs

Tab. 2 LEDs PERU

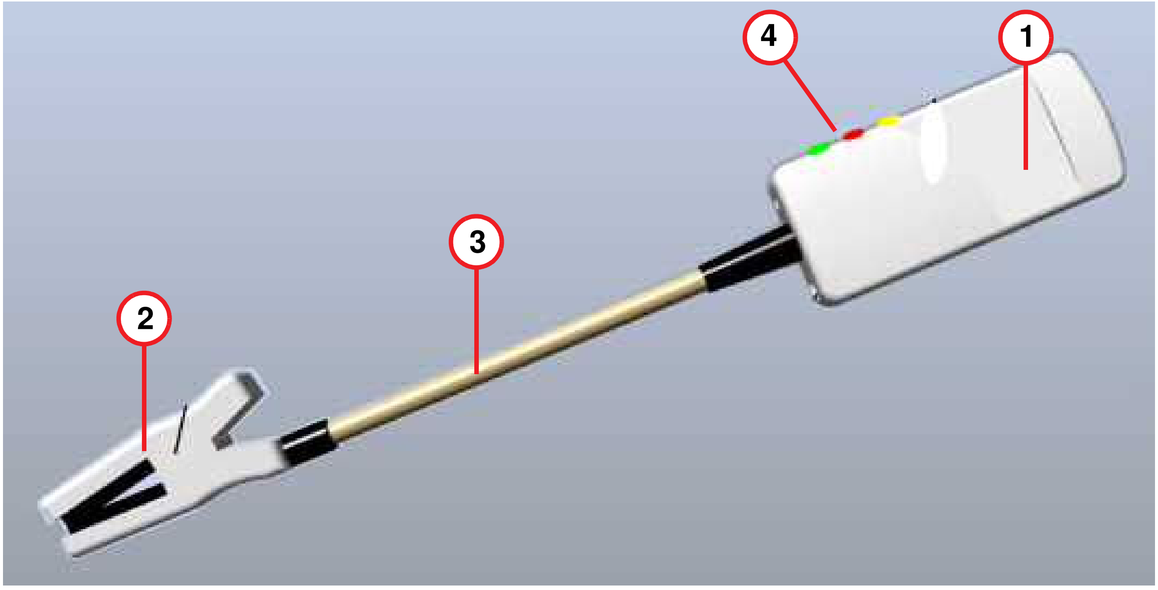
| LED | Description |
|---|---|
| Green | Battery status (if low, flashes twice in rapid succession) |
| Red | Electrode error (flashes if electrode is not correctly applied) |
| Yellow | Charging status (off if charging cycle is finished) |

### 4.2.3 Service routines

The PERU is equipped with a buil- in test signal generator that can be activated by starting the PMU interactive test in SESO. The AMC sends a test command over the CAN bus (LWL_INT) to the PDAU and then to the PERU via the wireless Bluetooth connection. An internal test signal (3.2 Hz) is connected to the preamplifier and routed back the same way to the AMC, from where it is displayed at the physiological window in syngo MR.

## 4.3 PPU

*Fig. 24: PPU*




(1) Transmitter
(2) Finger clip
(3) Fibre optic cable
(4) Status LEDs

### 4.3.1 Function

The PPU (Peripheral Pulse Unit) detects the peripheral pulse of a patient by means of a fiber optic finger clip. The signal is converted into an electrical pulse, digitized and transmitted via the second channel (2.474 GHz) of the Bluetooth wireless interface. Maximum transmit power is 1mW at 10% duty-cycle, usable transmission radius is about 3 meters.

The PPU is supplied by a integrated lithium battery that can be recharged in less than three hours by a wall-mounted charging station that is normally mounted close to the operator’s console.

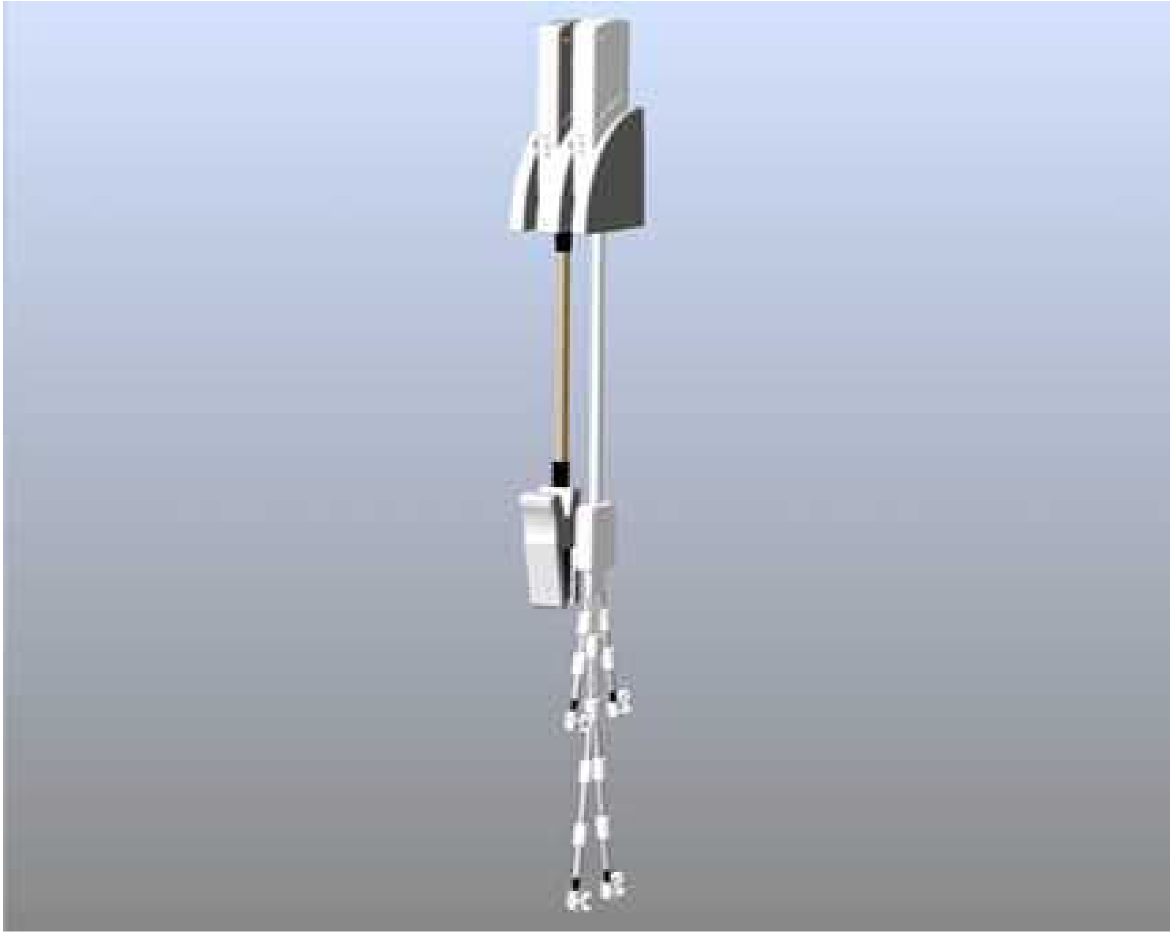
### 4.3.2 LEDs

Tab. 3 LEDs PPU

| LED | Description |
|---|---|
| Green | Battery status (if low, flashes twice in rapid succession) |
| Red | Sensor error (flashes if pulse sensor is not correctly applied) |
| Yellow | Charging status (off if charging cycle is finished) |

## 4.4 Battery charging station

*Fig. 25: Battery Charger*



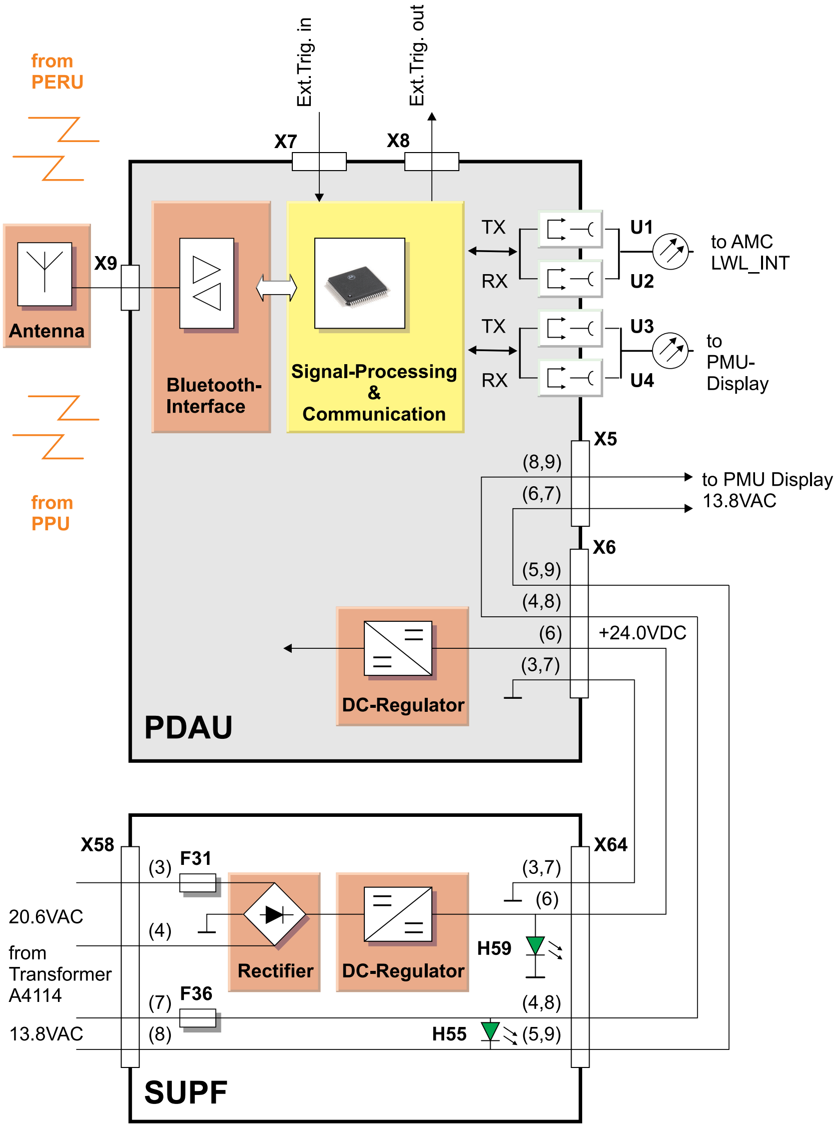
### 4.4.1 Function

A battery charger for PERU and PPU is delivered with the system. It can be wall-mounted and should be mounted close to the operator's console. The +5 VDC charging power is supplied by a cable connection from a free USB connector of the Host computer. Charging time for PERU and PPU battery is less than three hours. When the charging cycle is completed, the yellow control LEDs at PERU and PPU goes off.

If a PERU or PPU is stored for longer periods of time without being connected to the battery charger, it is recommended to connect the dummy plug (part of system delivery) to its charging connector to avoid battery discharging.

## 4.5 PDAU

*Fig. 26: PDAU*




### 4.5.1 Function

The PDAU (Physiological Data Acquisition Unit) receives the physiological signals from PERU and PPU, processes these and routes them via a bi-directional, serial, fiber-optic interface to the AMC and PMU Display. The unit is mounted on top of the Control electronics and is supplied with power by the SUPF connector X64.

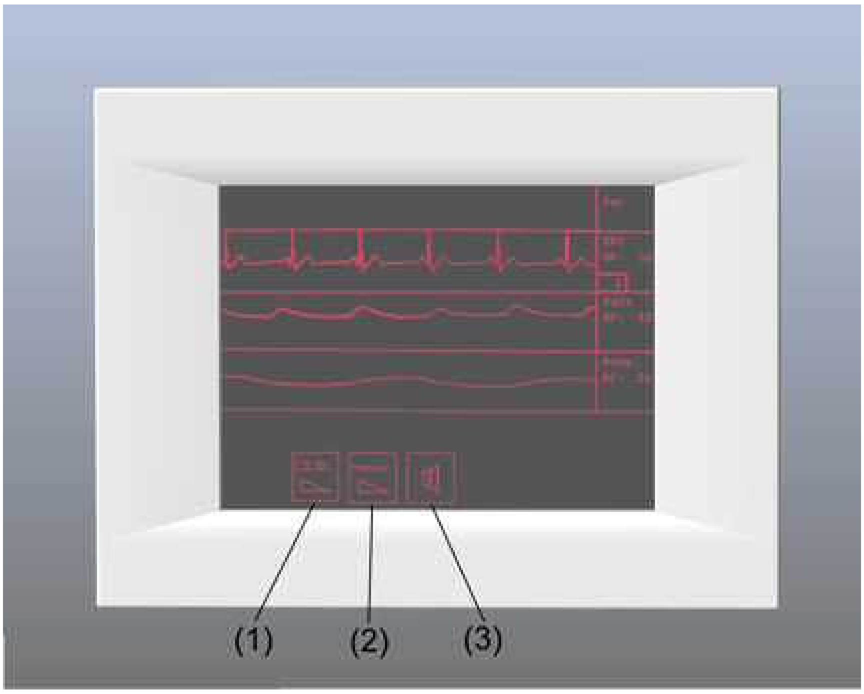
The receive-antenna for the physiological signals is located at the top right side behind the front funnel cover and connected via a coaxial cable to the PDAU.

For external triggering, the PDAU is equipped with two galvanically-isolated cinch connectors that are accessible at the left side of the lifting column cover. The TTL-trigger input can be used to trigger an MR sequence, the trigger output can be fed to an external device.

The PDAU supply raw voltage (+24.0 VDC at connector X6) is generated by the SUPF A4120, as well as the 13.8 VAC voltage supply for the PMU Display.

## 4.6 PMU Display (option)

*Fig. 27: PMU Display*




(1) ECG Channel Select
(2) Scroll Speed
(3) Acoustic Signal

### 4.6.1 Function

The PMU Display is able to display the physiological signal curves inside the examination room to facilitate the correct positioning of sensors to the patient. The PMU Display is mounted at the top left side of the magnet front cover.

A 13.8 VAC supply voltage for the PMU Display is derived by connector X5 at the PDAU. The bi-directional fiber-optic interface (FOC-connectors U3 and U4) supplies communication to the PDAU.

Four signals can be displayed:

- **ECG**
- **Pulse**
- **Respiratory signal**
- **External Triggering signal**

ECG, Pulse and Respiratory signal are displayed simultaneously in real time curves, the External Triggering signal is indicated as blinking "*" symbol for each low-to-high transition.

The PMU Display is equipped with three touchscreen buttons to adjust:

- **ECG Channel Select** (I or AVF for best R-wave signal)
- **Scroll Speed** (three speeds)

- **Acoustic Signal** (beeper activate/deactivate)

## 5.1 Overview

The Intercom system contains the following standard components:

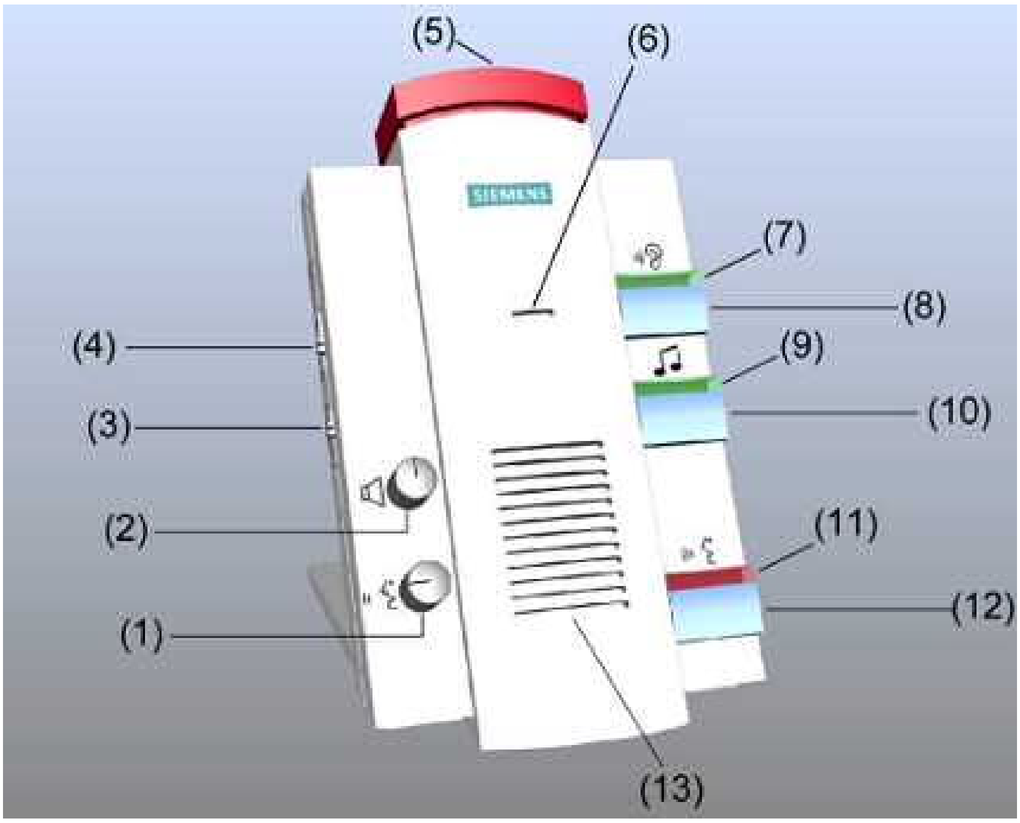
- **Central Unit Intercom** - This box is mounted in back of the operator's console. An external audio source, e.g. CD player, can be connected at "Music In" input to allow piping in music to the patient. R1 is a potentiometer to preset the max. amplification of the speech from the Control Room to the RF room. R2 is a potentiometer to preset the max. amplification of the speech from the RF room to the Control Room. Both potentiometers are factory preset to max. amplification (fully clockwise) for best audio quality. Change settings only if customer complains about unacceptable acoustic feedback and/or insufficient volume adjustment range at the Operating Unit.
- **Operating Unit Intercom** - Contains a red table stop button on top. Pressing button twice will stop a running sequence.
- **Microphone** - Located in magnet front cover.
- **Squeeze Bulb** - For nurse call. By pressing the bulb, an audible signal sounds on the Operating Unit Intercom.
- **Headphone** - The patient can hear operator announcements and music. Music is interrupted for announcements.
- **Loud Speaker** - Mounted on magnet cover, plays music and operator's announcements.

Additional options are:

- **CCD Camera** - In back of magnet. The image is transferred via fiber optic to the
- **Patient Monitor** - To adjust, see system documentation.

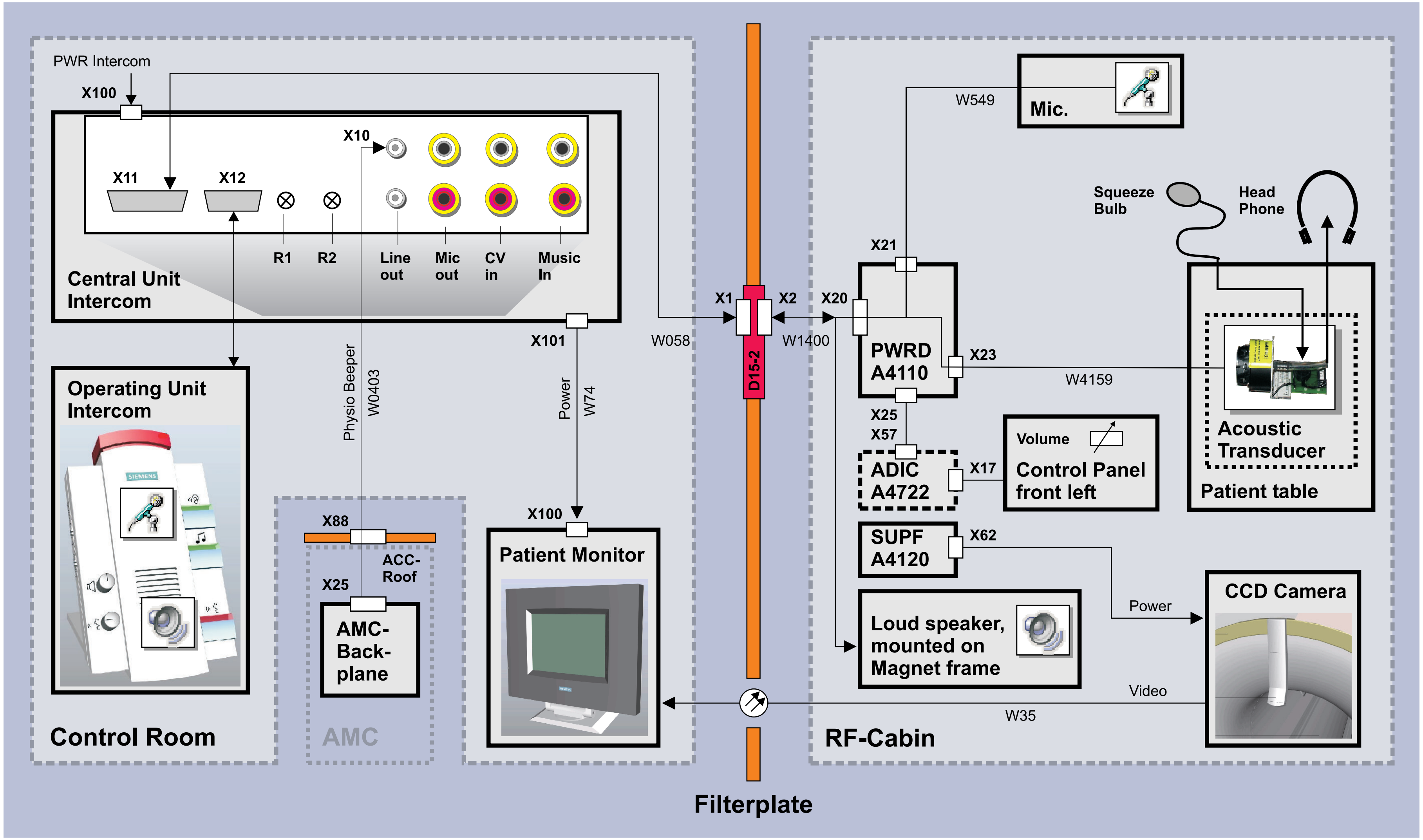
> i For intercom setup please refer to the Operating Instructions.

*Fig. 28: Operating Unit Intercom*



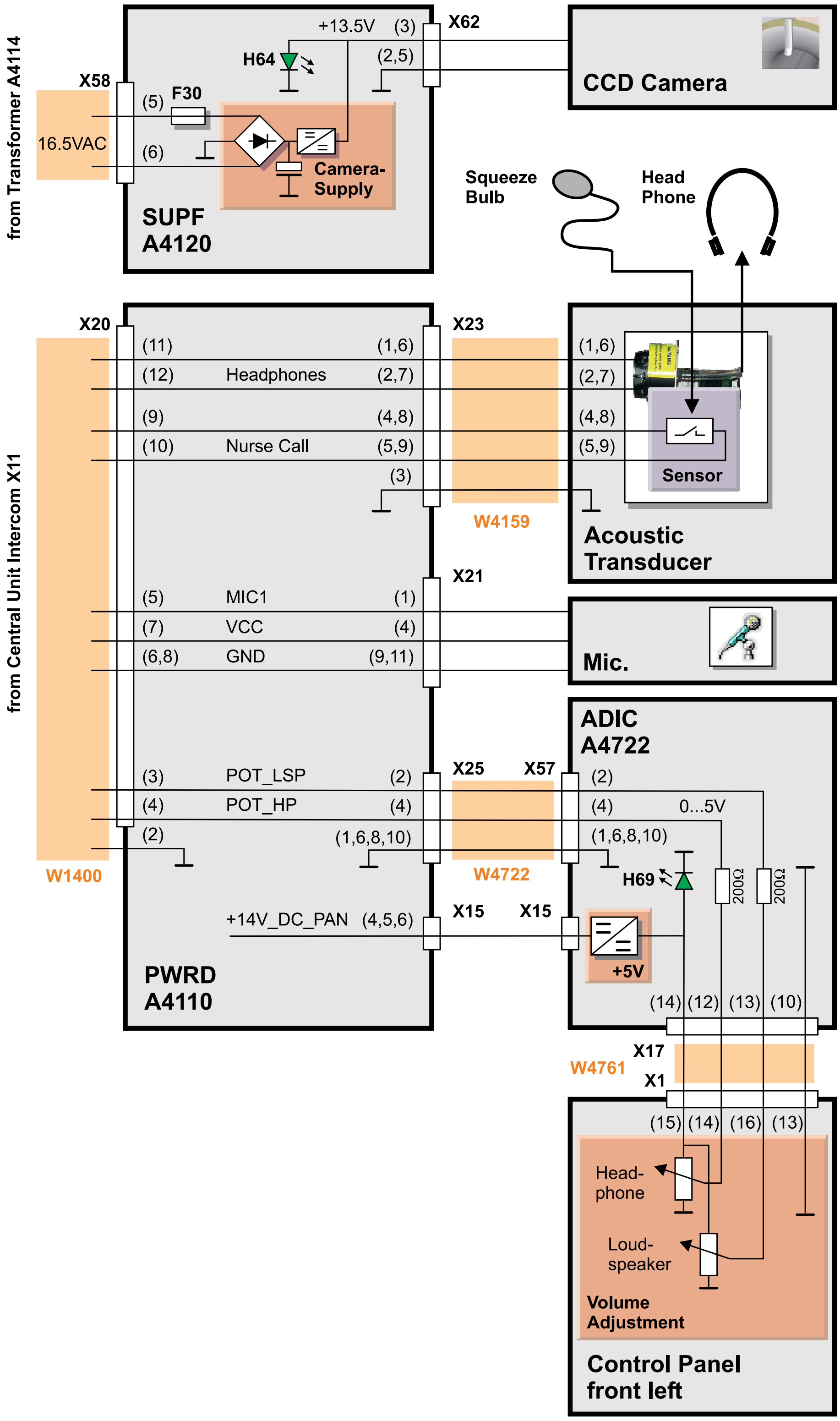

(1) Volume in RF cabin (speech only)
(2) Volume console (listening)
(3) CV/CBT switch (Computer Voice or background music)
(4) Physio beeper volume trigger (4 levels)
(5) Table stop/sequence stop button
(6) Microphone
(7) LED listening is on
(8) Button listening on/off
(9) LED music on
(10) Button play music in cabin on/off
(11) LED squeeze bulb active
(12) Speak button and squeeze bulb reset button
(13) Loudspeaker

*Fig. 29: Intercom*



## 5.2 Intercom connections

*Fig. 30: Intercom Connections*



- The common print number (M6-015.850.01) from Function Description is divided into separate Component print numbers.

There are no Hazard IDs in this document.

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